



# **KNOWLEDGE INSTITUTE OF TECHNOLOGY**

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**KAKAPALAYAM (PO), SALEM – 637 504**

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**LABORATORY MANUAL**

**FOR**

**CCS346 – EXPLORATORY DATA ANALYSIS LABORATORY**

## REGULATION 2021

### VISION, MISSION, PEOs AND PSOs OF CSE DEPARTMENT

| VISION                                                                                                                   |                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| To create globally competent software professionals with social values to cater the ever-changing industry requirements. |                                                                                                                                                      |
| MISSION                                                                                                                  |                                                                                                                                                      |
| M1                                                                                                                       | To provide appropriate infrastructure to impart need-based technical education through effective teaching and research                               |
| M2                                                                                                                       | To involve the students in collaborative projects on emerging technologies to fulfill the industrial requirements                                    |
| M3                                                                                                                       | To render value based education to students to take better engineering decision with social consciousness and to meet out the global standards       |
| M4                                                                                                                       | To inculcate leadership skills in students and encourage them to become a globally competent professional                                            |
| Programme Educational Objectives (PEOs)                                                                                  |                                                                                                                                                      |
| The graduates of Computer Science and Engineering will be able to                                                        |                                                                                                                                                      |
| PEO1                                                                                                                     | Pursue Higher Education and Research or have a successful career in industries associated with Computer Science and Engineering, or as Entrepreneurs |
| PEO2                                                                                                                     | Ensure that graduates will have the ability and attitude to adapt to emerging technological changes                                                  |
| PEO3                                                                                                                     | Acquire leadership skills to perform professional activities with social consciousness                                                               |
| Programme Specific Outcome (PSOs)                                                                                        |                                                                                                                                                      |
| The graduates will be able to                                                                                            |                                                                                                                                                      |
| PSO1                                                                                                                     | The students will be able to analyze large volume of data and make business decisions to improve efficiency with different algorithms and tools      |
| PSO2                                                                                                                     | The students will have the capacity to develop web and mobile applications for real time scenarios                                                   |
| PSO3                                                                                                                     | The students will be able to provide automation and smart solutions in various forms to the society with Internet of Things                          |

## **Course Code & Name: CCS346 & EXPLORATORY DATA ANALYSIS LABORATORY**

**REGULATION: R2021**

**YEAR/SEM: III/V**

### **COURSE OBJECTIVES:**

- To outline an overview of exploratory data analysis.
- To implement data visualization using Matplotlib.
- To perform univariate data exploration and analysis.
- To apply bivariate data exploration and analysis.
- To use Data exploration and visualization techniques for multivariate and time series data

### **LIST OF EXPERIMENTS:**

1. Install the data Analysis and Visualization tool: R/ Python /Tableau Public/ Power BI.
2. Perform exploratory data analysis (EDA) with datasets like email data set. Export all your emails as a dataset, import them inside a pandas data frame, visualize them and get different insights from the data.
3. Working with Numpy arrays, Pandas data frames , Basic plots using Matplotlib.
4. Explore various variable and row filters in R for cleaning data. Apply various plot features in R on sample data sets and visualize.
5. Perform Time Series Analysis and apply the various visualization techniques.
6. Perform Data Analysis and representation on a Map using various Map data sets with Mouse Rollover effect, user interaction, etc..
7. Build cartographic visualization for multiple datasets involving various countries of the world; states and districts in India etc.
8. Perform EDA on Wine Quality Data Set.
9. Use a case study on a data set and apply the various EDA and visualization techniques and present an analysis report.

### **COURSE OUTCOMES**

- Understand the fundamentals of exploratory data analysis.
- Implement the data visualization using Matplotlib.
- Perform univariate data exploration and analysis.
- Apply bivariate data exploration and analysis.
- Use Data exploration and visualization techniques for multivariate and time series data.

# CO's-PO's & PSO's MAPPING

| CO's | PO's |     |     |     |     |   |   |   |     |    |     |     | PSO's |     |     |
|------|------|-----|-----|-----|-----|---|---|---|-----|----|-----|-----|-------|-----|-----|
|      | 1    | 2   | 3   | 4   | 5   | 6 | 7 | 8 | 9   | 10 | 11  | 12  | 1     | 2   | 3   |
| 1    | 3    | 2   | 3   | 3   | 3   | - | - | - | 2   | 2  | 3   | 2   | 3     | 3   | 2   |
| 2    | 2    | 2   | 2   | 3   | 3   | - | - | - | 3   | 2  | 2   | 2   | 1     | 2   | 3   |
| 3    | 2    | 3   | 2   | 2   | 3   | - | - | - | 2   | 2  | 2   | 1   | 2     | 3   | 1   |
| 4    | 2    | 2   | 2   | 2   | 3   | - | - | - | 3   | 2  | 2   | 1   | 2     | 2   | 2   |
| 5    | 2    | 2   | 3   | 2   | 1   | - | - | - | 1   | 2  | 2   | 1   | 2     | 2   | 3   |
| AVg. | 2.2  | 2.2 | 2.4 | 2.4 | 2.6 | - | - | - | 2.2 | 2  | 2.2 | 1.4 | 2     | 2.4 | 2.2 |



## KNOWLEDGE INSTITUTE OF TECHNOLOGY

SALEM – 637 504

### DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

## CONTENTS

| Ex. No. | Date | Name of the Experiment                                                                                                                                                                                           | Page No. | Mark Awarded | Signature |
|---------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------|-----------|
| 1.      |      | Installation and Setup of Data Analysis and Visualization Tools: R, Python, Tableau, and Power BI.                                                                                                               |          |              |           |
| 2.      |      | Perform exploratory data analysis (EDA) with datasets like email data set. Export all your emails as a dataset, import them inside a pandas data frame, visualize them and get different insights from the data. |          |              |           |
| 3.      |      | Working with Numpy arrays, Pandas dataframe, Basic using Matplotlib.                                                                                                                                             |          |              |           |
| 4.      |      | Explore various variable and row filters in R for cleaning data. Apply various plot features in R on sample data sets and visualize.                                                                             |          |              |           |
| 5.      |      | Perform Time Series Analysis and apply the various visualization techniques.                                                                                                                                     |          |              |           |
| 6.      |      | Perform data analysis and representation on a map using various map datasets with mouse rollover effect, user interaction, etc.,.                                                                                |          |              |           |
| 7.      |      | Build cartographic visualization of multiple datasets involving various                                                                                                                                          |          |              |           |

|    |  |                                                                                                                      |  |  |  |
|----|--|----------------------------------------------------------------------------------------------------------------------|--|--|--|
|    |  | countries of the world; states, and districts in India                                                               |  |  |  |
| 8. |  | Perform EDA on Wine Quality Data Set.                                                                                |  |  |  |
| 9. |  | Use a case study on a dataset and apply the various EDA and visualization techniques and present an analysis report. |  |  |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Ex.no : 1</b><br><b>Date :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Installation and Setup of Data Analysis and Visualization Tools: R, Python, Tableau, and Power BI</b> |
| <p><b>Aim:</b></p> <p>The aim of this project is to install and set up four essential data analysis and visualization tools, namely R, Python, Tableau, and Power BI, to provide a robust environment for data analysis and visualization tasks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                          |
| <p><b>Algorithm:</b></p> <p>Step 1: System Requirements Assessment:</p> <ul style="list-style-type: none"> <li>- Before installation, ensure your system meets the minimum hardware and software requirements for each tool. Check the official documentation of R, Python, Tableau, and Power BI for system prerequisites.</li> </ul> <p>Step 2: Installation of R:</p> <ul style="list-style-type: none"> <li>- Download the R installation package from the official CRAN (Comprehensive R Archive Network) website (<a href="https://cran.r-project.org/mirrors.html">https://cran.r-project.org/mirrors.html</a>).</li> <li>- Follow the installation wizard, choose your preferred options, and install R on your system.</li> </ul> <p>Step 3: Installation of Python:</p> <ul style="list-style-type: none"> <li>- Download the latest Python installer from the official Python website (<a href="https://www.python.org/downloads/">https://www.python.org/downloads/</a>).</li> <li>- During installation, make sure to check the option "Add Python to PATH" for easy command-line access.</li> <li>- After installation, use pip (Python package manager) to install essential data science libraries, such as NumPy, Pandas, Matplotlib, and Jupyter Notebook, to enhance your data analysis capabilities.</li> </ul> <p>Step 4: Installation of Tableau:</p> <ul style="list-style-type: none"> <li>- Visit the official Tableau website (<a href="https://www.tableau.com/products/desktop/download">https://www.tableau.com/products/desktop/download</a>) to download the Tableau Desktop installer.</li> <li>- Follow the installation instructions and provide your licensing information or use the free trial period.</li> <li>- Ensure Tableau Desktop is successfully activated.</li> </ul> <p>Step 5: Installation of Power BI:</p> |                                                                                                          |

- Go to the Microsoft Power BI website (<https://powerbi.microsoft.com/en-us/desktop/>) to download Power BI Desktop.

- Download and install the Power BI Desktop application.

- You will need a Microsoft account to access certain features. Sign in or create an account if required.

#### Step 6: Configuration and Integration:

- Configure R and Python with data science IDEs such as RStudio or Jupyter Notebook to harness their full potential for data analysis.

- For Tableau and Power BI, explore their integration capabilities with databases, cloud services, and data sources you plan to use for analysis and visualization.

#### Step 7: Testing and Documentation:

- Verify that all the installed tools are working correctly by creating a simple data analysis and visualization project.

- Document the installation process, any challenges faced, and how they were overcome. This documentation will be helpful for future reference and troubleshooting.

## R:

### Step 1:



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## The R Project for Statistical Computing

### Getting Started

R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS. To [download R](#), please choose your preferred CRAN mirror.

If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

### News

- **R version 4.3.2 (Eye Holes) prerelease versions** will appear starting Saturday 2023-10-21. Final release is scheduled for Tuesday 2023-10-31.
- useR! 2024 will be a hybrid conference, taking place 8-11 July 2024 in Salzburg, Austria.
- **R version 4.3.1 (Beagle Scouts)** has been released on 2023-06-16.
- **R version 4.2.3 (Shortstop Beagle)** has been released on 2023-03-15.
- You can support the R Foundation with a renewable subscription as a [supporting member](#)

## Step 2:

0-Cloud

<https://cloud.r-project.org/>

Automatic redirection to servers worldwide, currently sponsored by Rstudio

## Step 3:

### The Comprehensive R Archive Network

#### Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

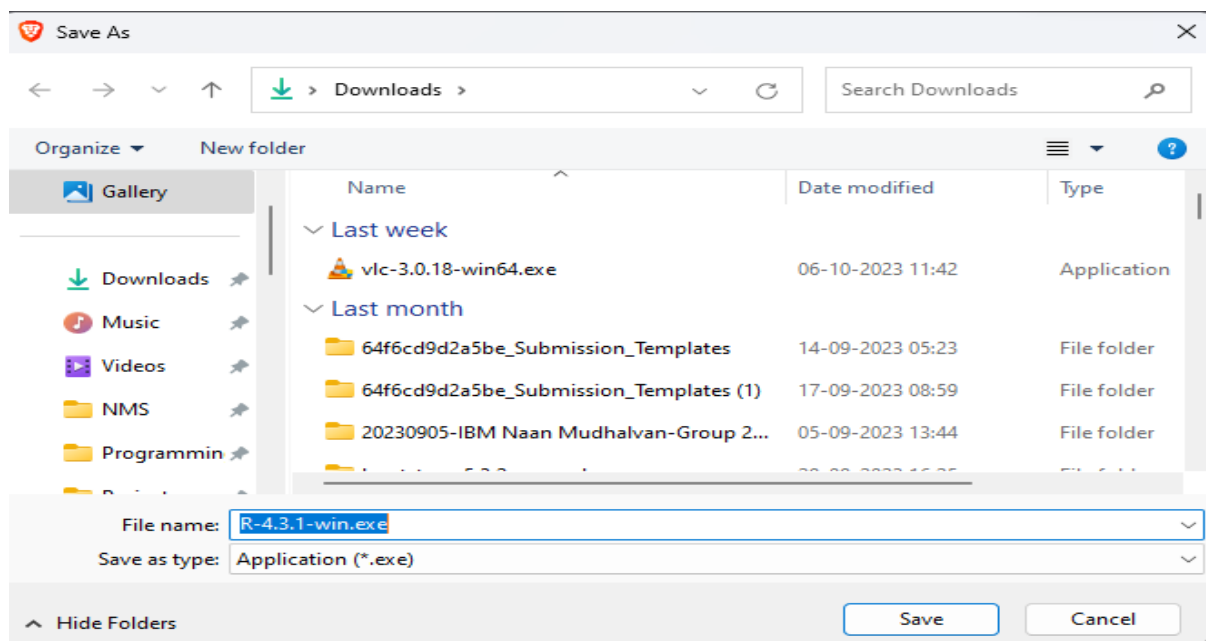
- [Download R for Linux](#) ([Debian](#), [Fedora/Redhat](#), [Ubuntu](#))
- [Download R for macOS](#)
- [Download R for Windows](#)

[Download R-4.3.1 for Windows](#) (79 megabytes, 64 bit)

[README on the Windows binary distribution](#)

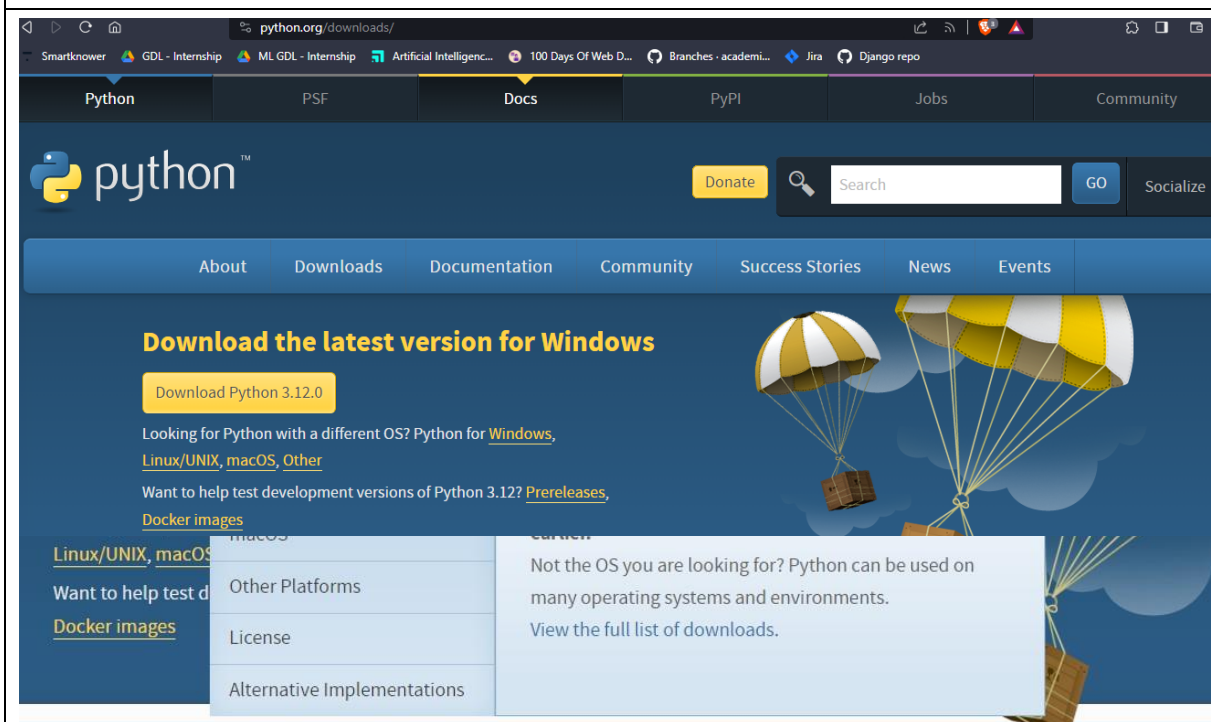
[New features in this version](#)

## Step 4:

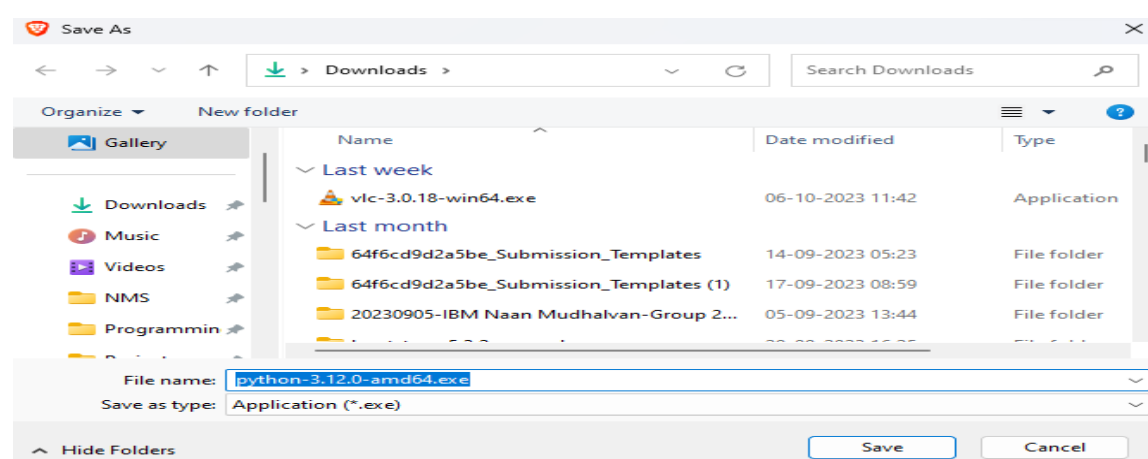


## Step 5:

Follow the next procedures to completely install R.



### Step 3:



### Step 4:

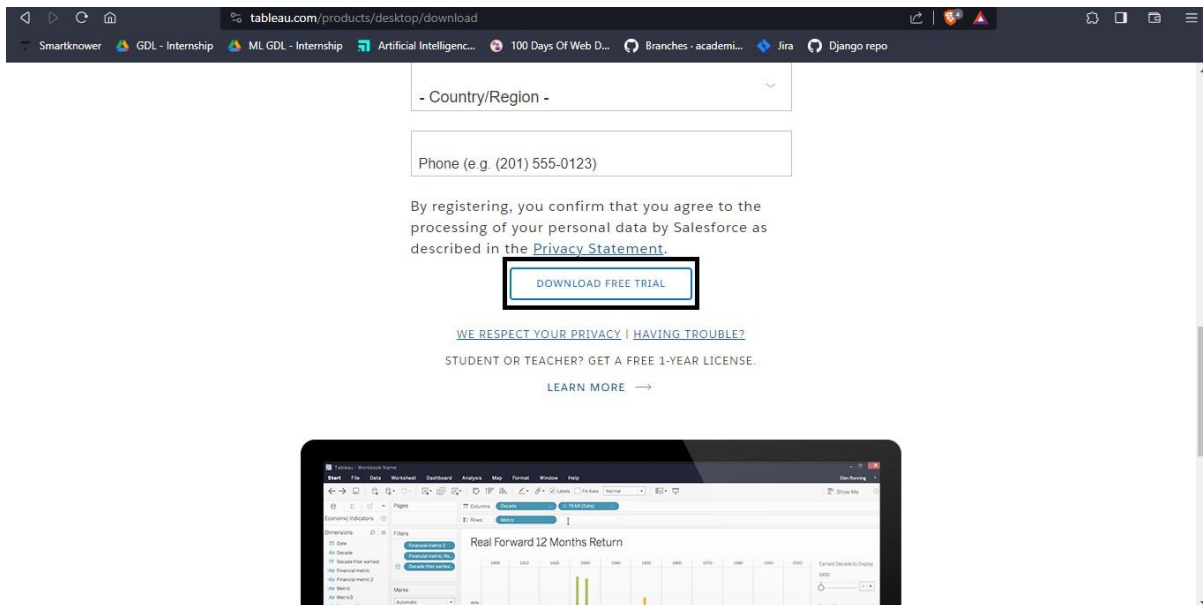
Follow the next procedures to completely install python.

## Tableau Public:

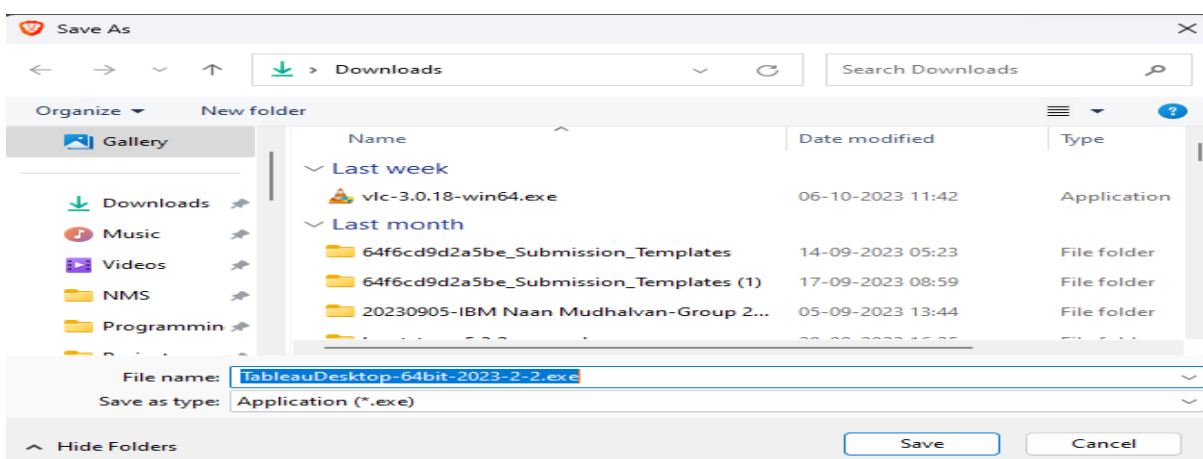
### Step 1:

Sign In to Tableau Public.

### Step 2:



### Step 3:

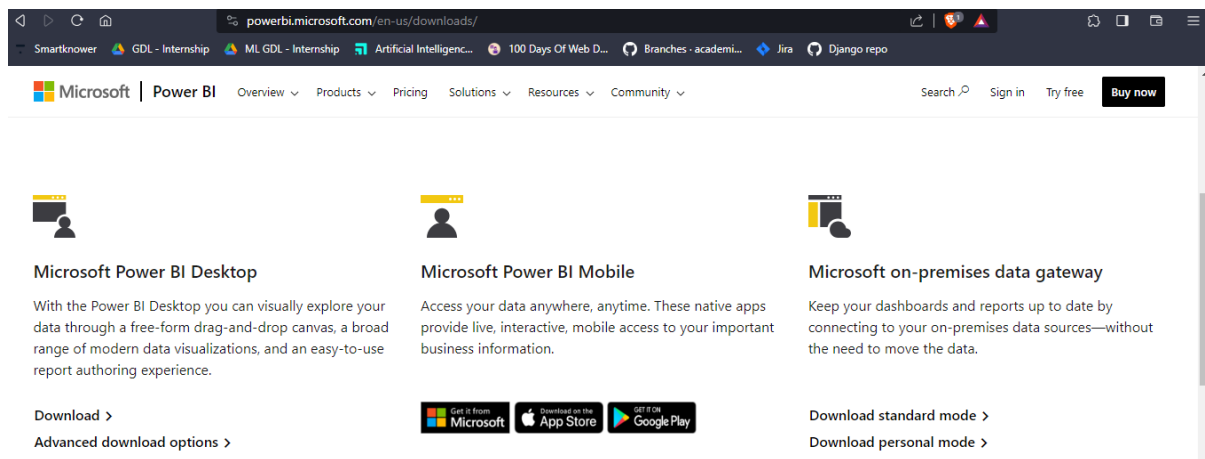


### Step 4:

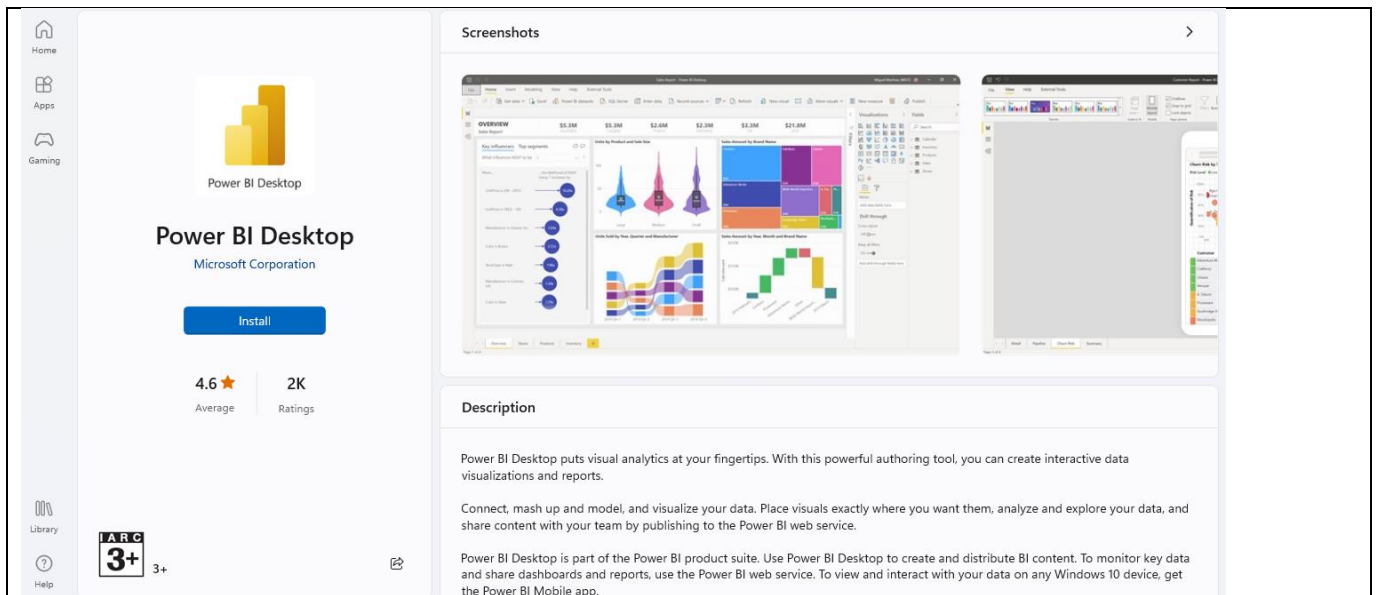
Follow the further instructions to install Tableau Public.

## Power BI:

### Step 1:



### Step 2:



### Step 3:

Follow the next procedures to completely install Power BI.

### Result:

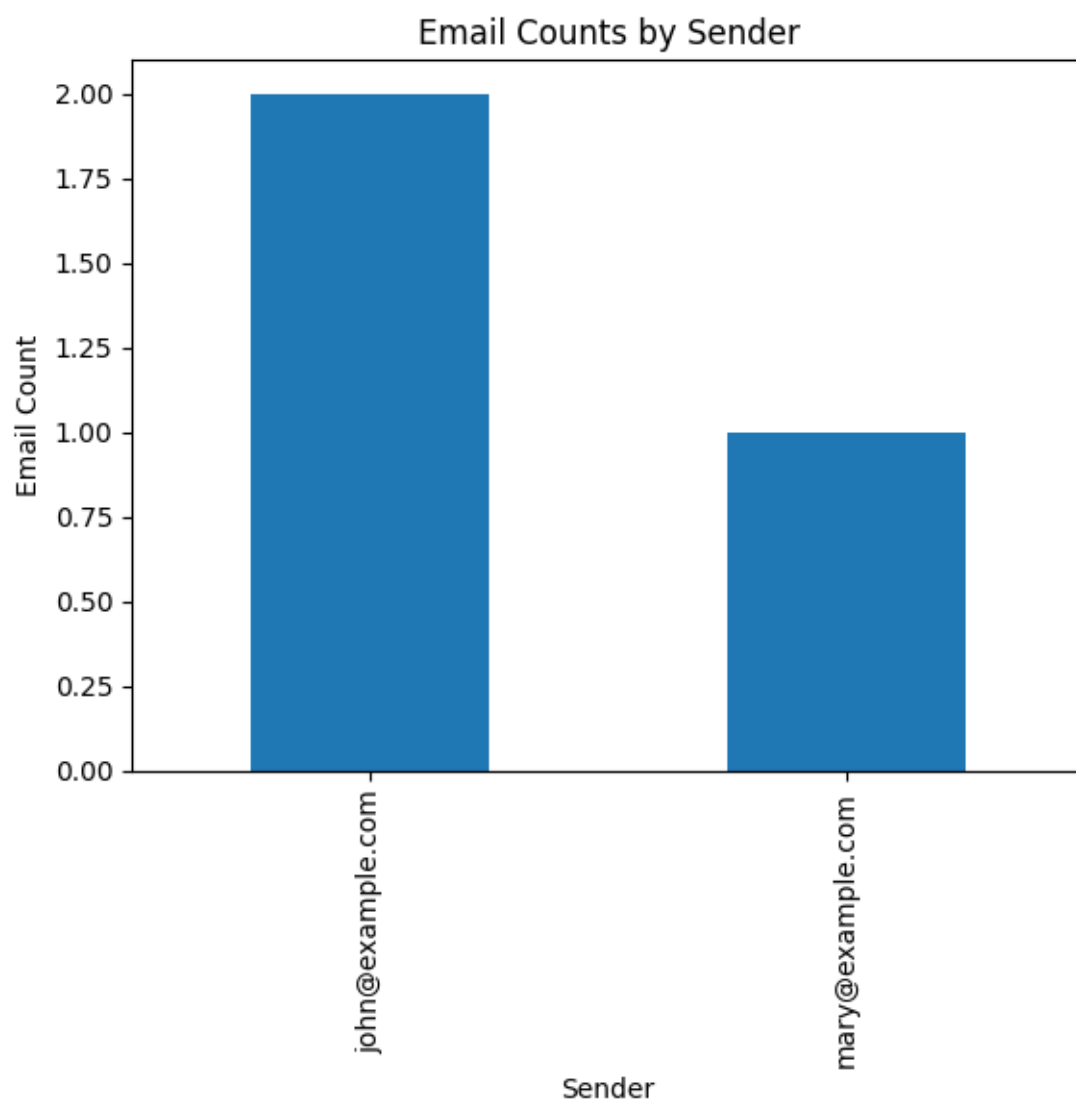
Thus the steps to install R/Python/Tableau Public/Power BI was completed successfully.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ex.no : 2</b><br><br><b>Date :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Perform exploratory data analysis (EDA) with datasets like email data set.</b><br><b>Export all your emails as a dataset, import them inside a pandas data frame, visualize them and get different insights from the data.</b> |
| <b>Aim :</b><br><br>Perform exploratory data analysis (EDA) with datasets like email data set. Export all your emails as a dataset, import them inside a pandas data frame, visualize them and get different insights from the data.                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                   |
| <b>Algorithm</b><br><br>Step 1: Load the data and create a DataFrame.<br><br>Step 2: Convert the 'timestamp' column to datetime format.<br><br>Step 3: Check the data's structure and data types.<br><br>Step 4: Preview the first few rows of the DataFrame.<br><br>Step 5: Find the most active email senders.<br><br>Step 6: Visualize common keywords in email content with a word cloud.<br><br>Step 7: Analyze email frequency over time using a time series plot.<br><br>Step 8: Explore additional insights based on specific requirements.<br><br>Step 9: Display results and provide interpretations.<br><br>Step 10: Conclude the analysis. |                                                                                                                                                                                                                                   |
| <b>Program :</b><br><br><pre>import pandas as pd  # Sample email data emails = {      'sender': ['john@example.com', 'mary@example.com', 'john@example.com'],     'receiver': ['mary@example.com', 'john@example.com', 'mary@example.com'],     'subject': ['Meeting Reminder', 'Regarding Project', 'Important Updates'],     'timestamp': ['2023-06-28 10:15:00', '2023-06-28 12:30:00', '2023-06-29 14:45:00'],     'content': ['Dear Mary, This is a reminder for our meeting tomorrow. Regards, John',                 'Hi John, I wanted to discuss the project with you. Let\'s connect. Regards, Mary',</pre>                                  |                                                                                                                                                                                                                                   |

| <pre>'Dear Mary, Please find attached the important updates. Best, John']  }</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                        |                                                        |                   |                     |           |   |   |                                                        |                                                        |                  |                     |  |   |                                                        |                                                        |                   |                     |  |   |                                                        |                                                        |                   |                     |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|-------------------|---------------------|-----------|---|---|--------------------------------------------------------|--------------------------------------------------------|------------------|---------------------|--|---|--------------------------------------------------------|--------------------------------------------------------|-------------------|---------------------|--|---|--------------------------------------------------------|--------------------------------------------------------|-------------------|---------------------|--|
| <pre># Create a DataFrame  df = pd.DataFrame(emails)</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                        |                                                        |                   |                     |           |   |   |                                                        |                                                        |                  |                     |  |   |                                                        |                                                        |                   |                     |  |   |                                                        |                                                        |                   |                     |  |
| <pre># Convert timestamp column to datetime  df['timestamp'] = pd.to_datetime(df['timestamp'])</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                        |                                                        |                   |                     |           |   |   |                                                        |                                                        |                  |                     |  |   |                                                        |                                                        |                   |                     |  |   |                                                        |                                                        |                   |                     |  |
| <pre># Print DataFrame information  print(df.info())</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                        |                                                        |                   |                     |           |   |   |                                                        |                                                        |                  |                     |  |   |                                                        |                                                        |                   |                     |  |   |                                                        |                                                        |                   |                     |  |
| <pre># Preview the first few rows  print(df.head())</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                        |                                                        |                   |                     |           |   |   |                                                        |                                                        |                  |                     |  |   |                                                        |                                                        |                   |                     |  |   |                                                        |                                                        |                   |                     |  |
| <p><b>Output</b></p> <pre>&lt;class 'pandas.core.frame.DataFrame'&gt; RangeIndex: 3 entries, 0 to 2 Data columns (total 5 columns):  #   Column      Non-Null Count  Dtype ---  -  0   sender      3 non-null      object  1   receiver    3 non-null      object  2   subject     3 non-null      object  3   timestamp   3 non-null      datetime64[ns]  4   content     3 non-null      object dtypes: datetime64[ns](1), object(4) memory usage: 248.0+ bytes None</pre> <table><tr><th></th><th>sender</th><th>receiver</th><th>subject</th><th>timestamp</th><th>\</th></tr><tr><td>0</td><td><a href="mailto:john@example.com">john@example.com</a></td><td><a href="mailto:mary@example.com">mary@example.com</a></td><td>Meeting Reminder</td><td>2023-06-28 10:15:00</td><td></td></tr><tr><td>1</td><td><a href="mailto:mary@example.com">mary@example.com</a></td><td><a href="mailto:john@example.com">john@example.com</a></td><td>Regarding Project</td><td>2023-06-28 12:30:00</td><td></td></tr><tr><td>2</td><td><a href="mailto:john@example.com">john@example.com</a></td><td><a href="mailto:mary@example.com">mary@example.com</a></td><td>Important Updates</td><td>2023-06-29 14:45:00</td><td></td></tr></table> <pre>                                 content 0   Dear Mary, This is a reminder for our meeting ... 1   Hi John, I wanted to discuss the project with ... 2   Dear Mary, Please find attached the important ...</pre> |                                                        | sender                                                 | receiver          | subject             | timestamp | \ | 0 | <a href="mailto:john@example.com">john@example.com</a> | <a href="mailto:mary@example.com">mary@example.com</a> | Meeting Reminder | 2023-06-28 10:15:00 |  | 1 | <a href="mailto:mary@example.com">mary@example.com</a> | <a href="mailto:john@example.com">john@example.com</a> | Regarding Project | 2023-06-28 12:30:00 |  | 2 | <a href="mailto:john@example.com">john@example.com</a> | <a href="mailto:mary@example.com">mary@example.com</a> | Important Updates | 2023-06-29 14:45:00 |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | sender                                                 | receiver                                               | subject           | timestamp           | \         |   |   |                                                        |                                                        |                  |                     |  |   |                                                        |                                                        |                   |                     |  |   |                                                        |                                                        |                   |                     |  |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <a href="mailto:john@example.com">john@example.com</a> | <a href="mailto:mary@example.com">mary@example.com</a> | Meeting Reminder  | 2023-06-28 10:15:00 |           |   |   |                                                        |                                                        |                  |                     |  |   |                                                        |                                                        |                   |                     |  |   |                                                        |                                                        |                   |                     |  |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <a href="mailto:mary@example.com">mary@example.com</a> | <a href="mailto:john@example.com">john@example.com</a> | Regarding Project | 2023-06-28 12:30:00 |           |   |   |                                                        |                                                        |                  |                     |  |   |                                                        |                                                        |                   |                     |  |   |                                                        |                                                        |                   |                     |  |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <a href="mailto:john@example.com">john@example.com</a> | <a href="mailto:mary@example.com">mary@example.com</a> | Important Updates | 2023-06-29 14:45:00 |           |   |   |                                                        |                                                        |                  |                     |  |   |                                                        |                                                        |                   |                     |  |   |                                                        |                                                        |                   |                     |  |
| <pre>import matplotlib.pyplot as plt  df['sender'].value_counts().plot(kind='bar')  plt.xlabel('Sender')</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                        |                                                        |                   |                     |           |   |   |                                                        |                                                        |                  |                     |  |   |                                                        |                                                        |                   |                     |  |   |                                                        |                                                        |                   |                     |  |

```
plt.ylabel('Email Count')  
  
plt.title('Email Counts by Sender')  
  
plt.show()
```

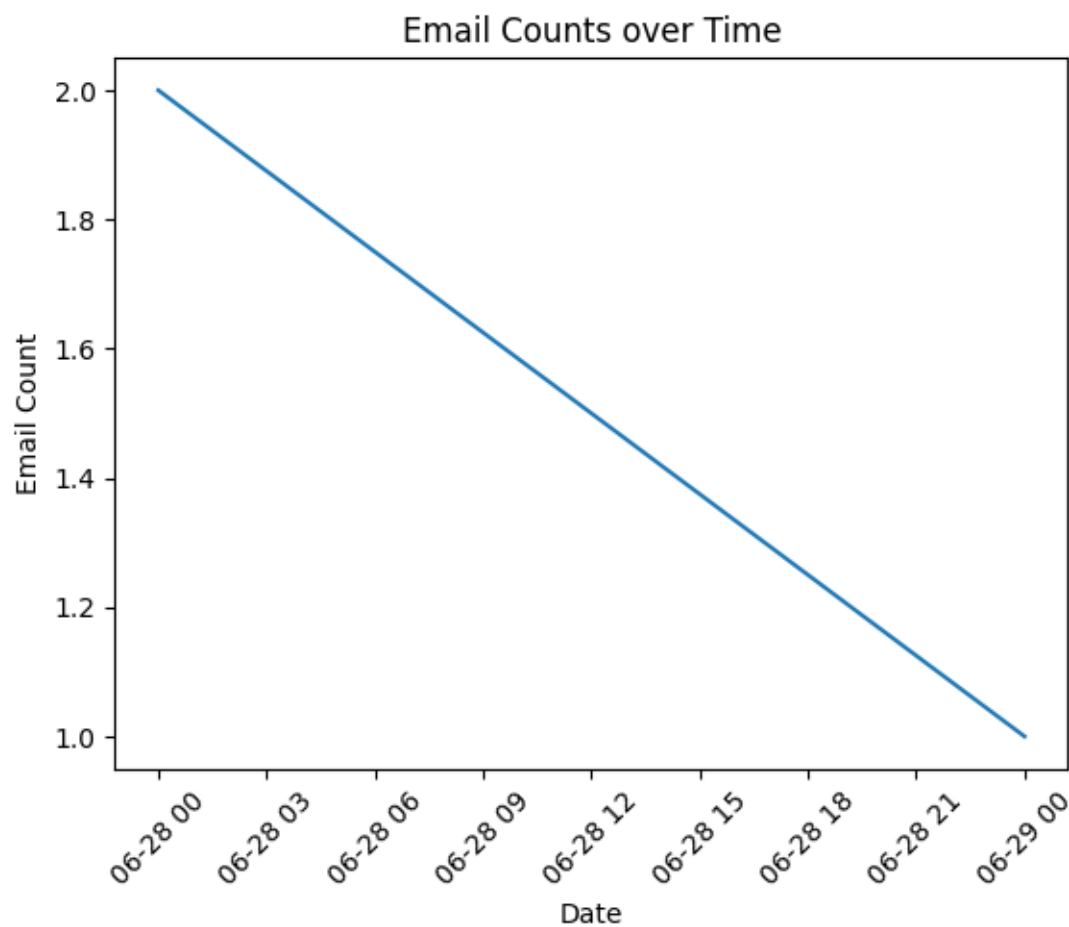
#### Output



```
df['timestamp'] = pd.to_datetime(df['timestamp'])  
  
df['date'] = df['timestamp'].dt.date  
  
email_counts = df['date'].value_counts().sort_index()  
  
plt.plot(email_counts.index, email_counts.values)
```

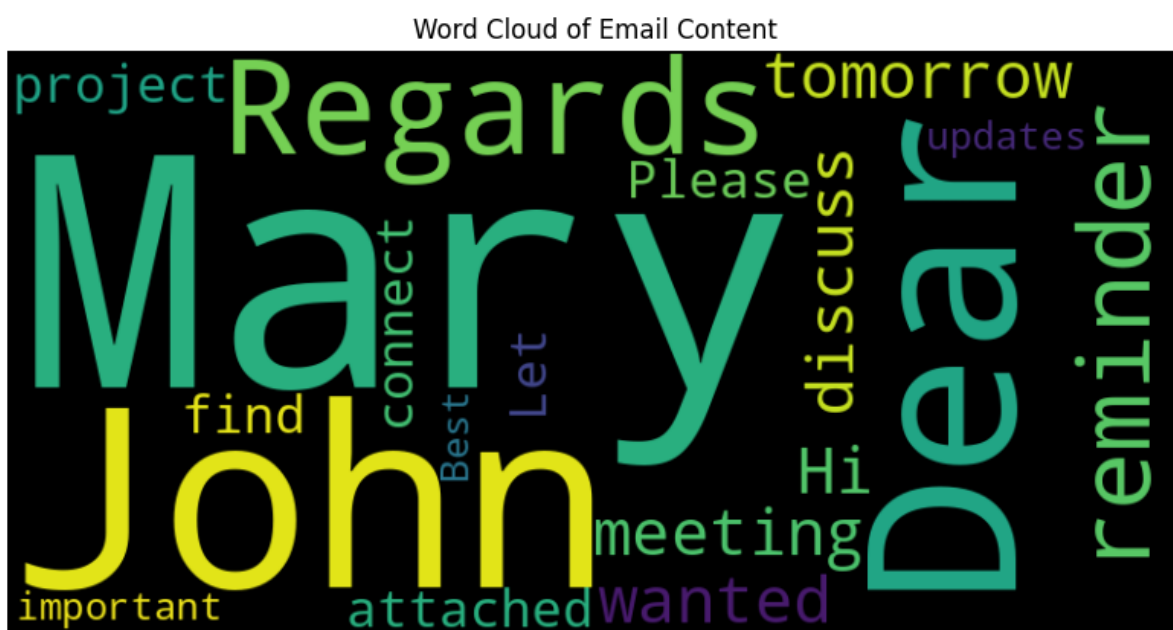
```
plt.xlabel('Date')
plt.ylabel('Email Count')
plt.title('Email Counts over Time')
plt.xticks(rotation=45)
plt.show()
```

### Output



```
plt.figure(figsize=(10, 5))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Word Cloud of Email Content')
plt.show()
```

## Output



```
most_active_senders = df['sender'].value_counts()
```

```
print(most_active_senders)
```

## Output

```
john@example.com      2  
mary@example.com      1  
Name: sender, dtype: int64
```

```
from wordcloud import WordCloud  
import matplotlib.pyplot as plt
```

```

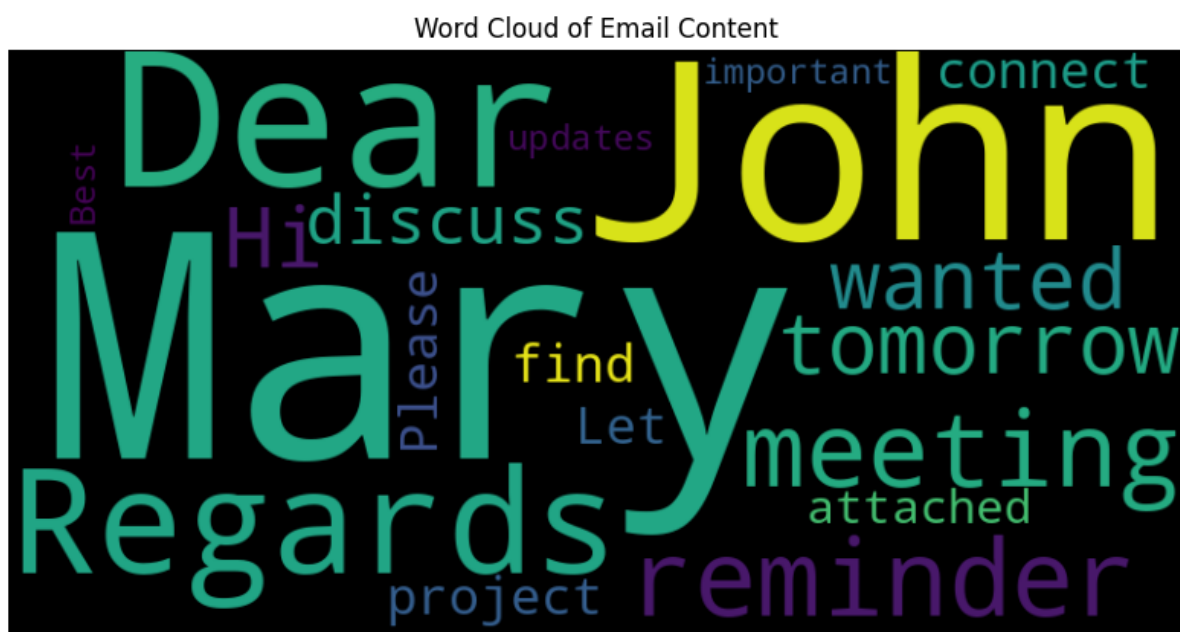
# Concatenate the email content
all_content = ' '.join(df['content'])

# Generate the word cloud
wordcloud = WordCloud(width=800, height=400).generate(all_content)

# Plot the word cloud
plt.figure(figsize=(10, 5))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Word Cloud of Email Content')
plt.show()

```

### Output



```

plt.plot(email_counts_per_day.index, email_counts_per_day.values)

plt.xlabel('Date')

plt.ylabel('Email Count')

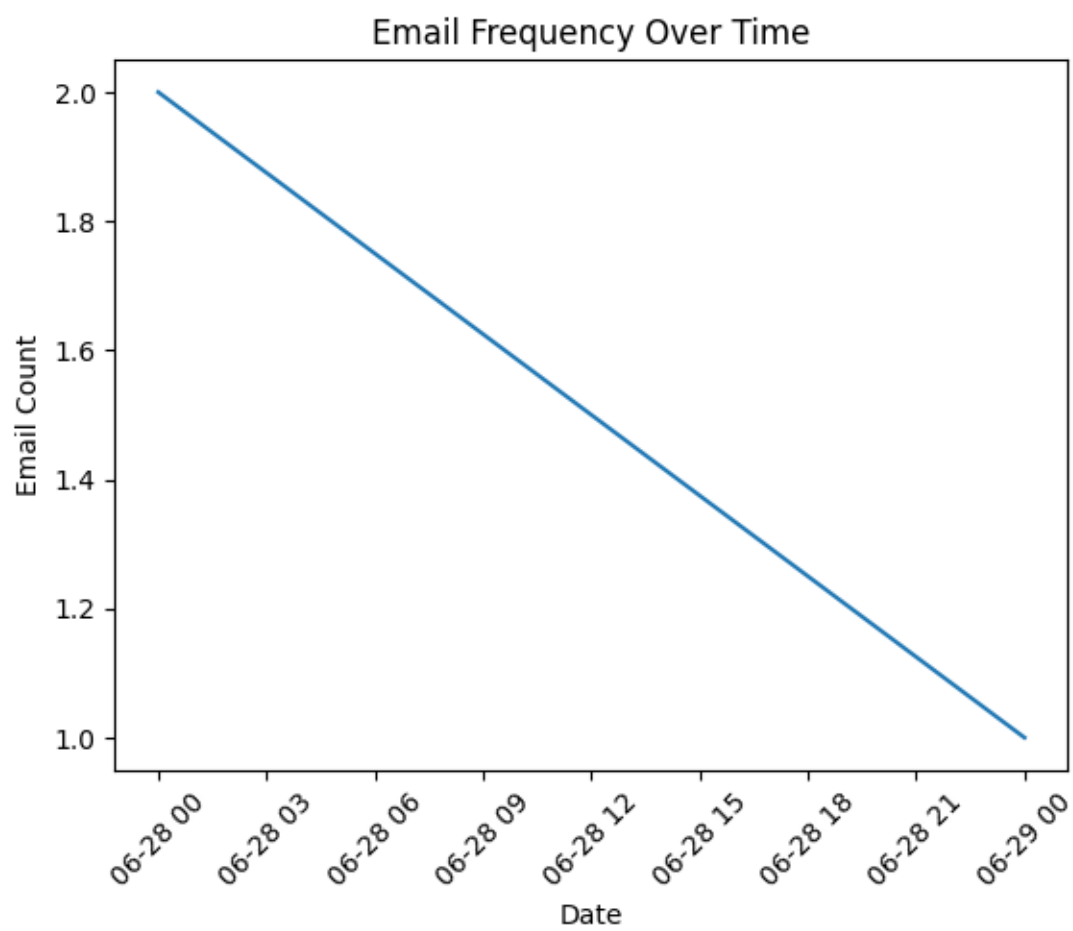
plt.title('Email Frequency Over Time')

plt.xticks(rotation=45)

plt.show()

```

**Output :**



|                                                                                   |
|-----------------------------------------------------------------------------------|
|                                                                                   |
| <b>Result :</b><br><br>Thus EDA on email dataset has been successfully performed. |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>EX NO: 3</b><br><br><b>DATE:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Working with Numpy arrays, Pandas dataframe, Basic using Matplotlib</b> |
| <b>Aim</b><br><br>To work with arrays in numpy module, dataframe with pandas module and basic plots with matplotlib module in python programming.                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                            |
| <b>Algorithm for Numpy</b><br><br>Step 1: Start the program<br><br>Step 2: Import the required packages<br><br>Step 3: Create the 1-d array, 2-d array by using built-in methods<br><br>Step 4: Generate arrays using zeros, ones, arange and linspace.<br><br>Step 5: Check the number of dimensions, size of an array<br><br>Step 6: Compute the shape of an array and reshape an array and perform transpose of an array<br><br>Step 7: Do the required operations like slicing, iterating and splitting an array element.<br><br>Step 8: Stop the program |                                                                            |
| <b><u>Numpy arrays</u></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                            |

### **(i) Creating numpy 1d and 2d arrays**

#### **Program**

```
import numpy as np

arr = np.array([1, 2, 3, 4, 5, 6])

a = np.array([[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12]])

print("1-d array:",arr)

print("2-d array:")

print(a)
```

#### **Output**

1-d array: [1 2 3 4 5 6]

2-d array:

[[ 1 2 3 4]

[ 5 6 7 8]

[ 9 10 11 12]]

### **(ii) Different ways of creating arrays**

#### **Program**

```
zer=np.zeros(2)

on=np.ones(3)

np.arange(2, 9, 2)

np.linspace(0, 10, num=5, dtype= np.int64)

print(zer)

print(on)
```

#### **Output**

array([0. 0.])

array([1., 1., 1.])

array([2, 4, 6, 8])

```
array([ 0,  2,  5,  7, 10], dtype=int64)
```

### **(iii) Numpy array Dimension,Shape,Size,Transpose and Reshaping**

#### **Program**

```
nums = np.array([[[1, 5, 2, 1],  
                  [4, 3, 5, 6],  
                  [6, 3, 0, 6],  
                  [7, 3, 5, 0],  
                  [2, 3, 3, 5]],  
                [[2, 2, 3, 1],  
                 [4, 0, 0, 5],  
                 [6, 3, 2, 1],  
                 [5, 1, 0, 0],  
                 [0, 1, 9, 1]],  
                [[3, 1, 4, 2],  
                 [4, 1, 6, 0],  
                 [1, 2, 0, 6],  
                 [8, 3, 4, 0],  
                 [2, 0, 2, 8]]])  
  
print("Number of dimensions :",nums.ndim)  
print("Size :",nums.size)  
print("Shape of array :",nums.shape)  
print("Transpose of array:",nums.T)  
print("Array after reshaping :",nums.reshape(3,4,5))
```

#### **Output**

Number of dimensions : 3

Size : 60

Shape of array : (3, 5, 4)

Transpose of array:

[[[1 2 3]

[4 4 4]

[6 6 1]

[7 5 8]

[2 0 2]]

[[[5 2 1]

[3 0 1]

[3 3 2]

[3 1 3]

[3 1 0]]

[[[2 3 4]

[5 0 6]

[0 2 0]

[5 0 4]

[3 9 2]]

[[[1 1 2]

[6 5 0]

[6 1 6]

[0 0 0]

[5 1 8]]]

Array after reshaping :

[[[1 5 2 1 4]

[3 5 6 6 3]

[0 6 7 3 5]

[0 2 3 3 5]]

```
[[2 2 3 1 4]
 [0 0 5 6 3]
 [2 1 5 1 0]
 [0 0 1 9 1]]
[[3 1 4 2 4]
 [1 6 0 1 2]
 [0 6 8 3 4]
 [0 2 0 2 8]]]
```

#### **(iv) Slicing and accessing arrays**

##### **Program**

```
p = np.array([1,2,3,4,5,6])
print(arr[2] + arr[3])
p[1:4]
```

##### **Output**

```
7
array([2, 3, 4])
```

#### **(v) Iterating Arrays**

##### **Program**

```
import numpy as np
array1 = np.array([1, 2, 3])
for x in array1:
    print(x)
```

##### **Output**

```
1
2
3
```

#### **(vi) Vstack,Hstack,split and flip functions on arrays**

##### **Program**

```

arr1 = np.array([[1, 1],
                 [2, 2]])
arr2 = np.array([[3, 3],
                 [4, 4]])
print("Vertical Stacking :",np.vstack((arr1,arr2)))
print("Horizontal stacking :",np.hstack((arr1,arr2)))
print("Splitting of array :",np.array_split(arr1,3))
print("Reversing of array :",np.flip(arr1))

```

### Output

Vertical Stacking :

```

array([[1, 1],
       [2, 2],
       [3, 3],
       [4, 4]])

```

Horizontal stacking :

```

array([[1, 1, 3, 3],
       [2, 2, 4, 4]])

```

Splitting of array :

```

[array([[1, 1]]), array([[2, 2]]), array([], shape=(0, 2), dtype=int32)]

```

Reversing of array :

```

array([[2, 2],
       [1, 1]])

```

### (vii) Operations on array

#### Program

```

b = np.array([[0.45053314, 0.17296777, 0.34376245, 0.5510652],
              [0.54627315, 0.05093587, 0.40067661, 0.55645993],
              [0.12697628, 0.82485143, 0.26590556, 0.56917101]])

```

```
print("maximum :",b.max())  
  
print("mean :",b.mean())  
  
print("standard deviation :",b.std())
```

### **Output**

```
maximum : 0.82485143  
  
mean : 0.40496486666666667  
  
standard deviation : 0.21392120766089617
```

### **Algorithm for Pandas:**

Step 1: Start the program

Step 2: Import the required packages

Step 3: Create a DataFrame using built-in methods and add a named index to it

Step 4: Load data into a DataFrame object otherwise Load Files(excel/csv) into a DataFrame

Step 5: Display the information about the data using built-in method

Step 6: Display the first five rows and last five rows using head and tail method

Step 7: Do the operation like slicing by different slicing methods

Step 8: Stop the program

### **Pandas DataFrame**

#### **(i) Creating Pandas DataFrame**

##### **Program**

```
import pandas as pd  
  
data = {  
  
"Total marks": [420, 380, 390],  
  
"study hours": [5, 4, 4]  
  
}  
  
dataframe = pd.DataFrame(data)  
  
print(dataframe)
```

**Output**

|   | Total marks | study hours |
|---|-------------|-------------|
| 0 | 420         | 5           |
| 1 | 380         | 4           |
| 2 | 390         | 4           |

**(ii) Creating named index****Program**

```
dataframe1=pd.DataFrame(data,index=['a','b','c'])
```

```
dataframe1
```

**Output**

|   | Total marks | study hours |
|---|-------------|-------------|
| a | 420         | 5           |
| b | 380         | 4           |
| c | 390         | 4           |

**(iii) Importing and adding into DataFrame****Program**

```
d=pd.read_csv("C:\\Users\\ADMIN\\Downloads\\winequality-red.csv")
```

```
#loading files into dataframe
```

```
df=pd.DataFrame(d)
```

```
df
```

**Output**

|      | fixed acidity | volatile acidity | citric acid | residual sugar | chlorides | free sulfur dioxide | total sulfur dioxide | density | pH   | sulphates | alcohol | quality |
|------|---------------|------------------|-------------|----------------|-----------|---------------------|----------------------|---------|------|-----------|---------|---------|
| 0    | 7.4           | 0.700            | 0.00        | 1.9            | 0.076     | 11.0                | 34.0                 | 0.99780 | 3.51 | 0.56      | 9.4     | 5       |
| 1    | 7.8           | 0.880            | 0.00        | 2.6            | 0.098     | 25.0                | 67.0                 | 0.99680 | 3.20 | 0.68      | 9.8     | 5       |
| 2    | 7.8           | 0.760            | 0.04        | 2.3            | 0.092     | 15.0                | 54.0                 | 0.99700 | 3.26 | 0.65      | 9.8     | 5       |
| 3    | 11.2          | 0.280            | 0.56        | 1.9            | 0.075     | 17.0                | 60.0                 | 0.99800 | 3.16 | 0.58      | 9.8     | 6       |
| 4    | 7.4           | 0.700            | 0.00        | 1.9            | 0.076     | 11.0                | 34.0                 | 0.99780 | 3.51 | 0.56      | 9.4     | 5       |
| ...  | ...           | ...              | ...         | ...            | ...       | ...                 | ...                  | ...     | ...  | ...       | ...     | ...     |
| 1594 | 6.2           | 0.600            | 0.08        | 2.0            | 0.090     | 32.0                | 44.0                 | 0.99490 | 3.45 | 0.58      | 10.5    | 5       |
| 1595 | 5.9           | 0.550            | 0.10        | 2.2            | 0.062     | 39.0                | 51.0                 | 0.99512 | 3.52 | 0.76      | 11.2    | 6       |
| 1596 | 6.3           | 0.510            | 0.13        | 2.3            | 0.076     | 29.0                | 40.0                 | 0.99574 | 3.42 | 0.75      | 11.0    | 6       |
| 1597 | 5.9           | 0.645            | 0.12        | 2.0            | 0.075     | 32.0                | 44.0                 | 0.99547 | 3.57 | 0.71      | 10.2    | 5       |
| 1598 | 6.0           | 0.310            | 0.47        | 3.6            | 0.067     | 18.0                | 42.0                 | 0.99549 | 3.39 | 0.66      | 11.0    | 6       |

1599 rows × 12 columns

#### (iv) Information about dataset

##### Program

```
df.info()
```

##### Output

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1599 entries, 0 to 1598
```

```
Data columns (total 12 columns):
```

```
#   Column                Non-Null Count  Dtype
---  -
0   fixed acidity          1599 non-null  float64
1   volatile acidity        1599 non-null  float64
2   citric acid             1599 non-null  float64
3   residual sugar          1599 non-null  float64
4   chlorides               1599 non-null  float64
5   free sulfur dioxide      1599 non-null  float64
6   total sulfur dioxide     1599 non-null  float64
7   density                 1599 non-null  float64
8   pH                     1599 non-null  float64
```

9

sulphates

1599 non-null

float64

10

alcohol

1599 non-null

float64

11

quality

1599 non-null

int64

dtypes: float64(11), int64(1)

memory usage: 150.0 KB

(v) Basic operations on DataFrame

Program

df.head()

df.tail()

Output

|      | fixed acidity | volatile acidity | citric acid | residual sugar | chlorides | free sulfur dioxide | total sulfur dioxide | density | pH   | sulphates | alcohol | quality |
|------|---------------|------------------|-------------|----------------|-----------|---------------------|----------------------|---------|------|-----------|---------|---------|
| 0    | 7.4           | 0.70             | 0.00        | 1.9            | 0.076     | 11.0                | 34.0                 | 0.9978  | 3.51 | 0.56      | 9.4     | 5       |
| 1    | 7.8           | 0.88             | 0.00        | 2.6            | 0.098     | 25.0                | 67.0                 | 0.9968  | 3.20 | 0.68      | 9.8     | 5       |
| 2    | 7.8           | 0.76             | 0.04        | 2.3            | 0.092     | 15.0                | 54.0                 | 0.9970  | 3.26 | 0.65      | 9.8     | 5       |
| 3    | 11.2          | 0.28             | 0.56        | 1.9            | 0.075     | 17.0                | 60.0                 | 0.9980  | 3.16 | 0.58      | 9.8     | 6       |
| 4    | 7.4           | 0.70             | 0.00        | 1.9            | 0.076     | 11.0                | 34.0                 | 0.9978  | 3.51 | 0.56      | 9.4     | 5       |
| :    | fixed acidity | volatile acidity | citric acid | residual sugar | chlorides | free sulfur dioxide | total sulfur dioxide | density | pH   | sulphates | alcohol | quality |
| 1594 | 6.2           | 0.600            | 0.08        | 2.0            | 0.090     | 32.0                | 44.0                 | 0.99490 | 3.45 | 0.58      | 10.5    | 5       |
| 1595 | 5.9           | 0.550            | 0.10        | 2.2            | 0.062     | 39.0                | 51.0                 | 0.99512 | 3.52 | 0.76      | 11.2    | 6       |
| 1596 | 6.3           | 0.510            | 0.13        | 2.3            | 0.076     | 29.0                | 40.0                 | 0.99574 | 3.42 | 0.75      | 11.0    | 6       |
| 1597 | 5.9           | 0.645            | 0.12        | 2.0            | 0.075     | 32.0                | 44.0                 | 0.99547 | 3.57 | 0.71      | 10.2    | 5       |
| 1598 | 6.0           | 0.310            | 0.47        | 3.6            | 0.067     | 18.0                | 42.0                 | 0.99549 | 3.39 | 0.66      | 11.0    | 6       |

(VI) Slicing using loc and iloc

Program

df.loc[200:220, ["residual sugar", "quality"]]

Output



|     | residual sugar | quality |
|-----|----------------|---------|
| 200 | 1.4            | 7       |
| 201 | 2.1            | 5       |
| 202 | 1.5            | 5       |
| 203 | 1.6            | 5       |
| 204 | 1.6            | 6       |
| 205 | 2.6            | 7       |
| 206 | 2.6            | 7       |
| 207 | 1.8            | 5       |
| 208 | 2.7            | 5       |
| 209 | 2.1            | 7       |
| 210 | 2.0            | 6       |
| 211 | 2.8            | 6       |
| 212 | 2.1            | 6       |
| 213 | 2.2            | 5       |
| 214 | 2.4            | 6       |
| 215 | 5.6            | 5       |
| 216 | 2.0            | 5       |
| 217 | 2.2            | 5       |
| 218 | 1.9            | 5       |
| 219 | 2.4            | 5       |
| 220 | 2.0            | 6       |

**Program**  

```
df.iloc[3:16, 0:7]
```

**Output**

|    | fixed acidity | volatile acidity | citric acid | residual sugar | chlorides | free sulfur dioxide | total sulfur dioxide |
|----|---------------|------------------|-------------|----------------|-----------|---------------------|----------------------|
| 3  | 11.2          | 0.280            | 0.56        | 1.9            | 0.075     | 17.0                | 60.0                 |
| 4  | 7.4           | 0.700            | 0.00        | 1.9            | 0.076     | 11.0                | 34.0                 |
| 5  | 7.4           | 0.660            | 0.00        | 1.8            | 0.075     | 13.0                | 40.0                 |
| 6  | 7.9           | 0.600            | 0.06        | 1.6            | 0.069     | 15.0                | 59.0                 |
| 7  | 7.3           | 0.650            | 0.00        | 1.2            | 0.065     | 15.0                | 21.0                 |
| 8  | 7.8           | 0.580            | 0.02        | 2.0            | 0.073     | 9.0                 | 18.0                 |
| 9  | 7.5           | 0.500            | 0.36        | 6.1            | 0.071     | 17.0                | 102.0                |
| 10 | 6.7           | 0.580            | 0.08        | 1.8            | 0.097     | 15.0                | 65.0                 |
| 11 | 7.5           | 0.500            | 0.36        | 6.1            | 0.071     | 17.0                | 102.0                |
| 12 | 5.6           | 0.615            | 0.00        | 1.6            | 0.089     | 16.0                | 59.0                 |
| 13 | 7.8           | 0.610            | 0.29        | 1.6            | 0.114     | 9.0                 | 29.0                 |
| 14 | 8.9           | 0.620            | 0.18        | 3.8            | 0.176     | 52.0                | 145.0                |
| 15 | 8.9           | 0.620            | 0.19        | 3.9            | 0.170     | 51.0                | 148.0                |

### **Algorithm for Matplotlib:**

Step 1: Start the program

Step 2: Import the required packages

Step 3: Visualize the data using simple line plot

Step 4: Create a bar plot, use the bar method and customize the appearance, labels, and title.

Step 5: Generate a scatter plot by scatter function to plot your data points and customize the size, color, and style of the markers, as well as labels and title.

Step 6: Visualize the data using the pie chart and histogram with required parameters

Step 7: Add grid line to the plots by using built-in methods

Step 8: Stop the program

### **Basic plots using Matplotlib**

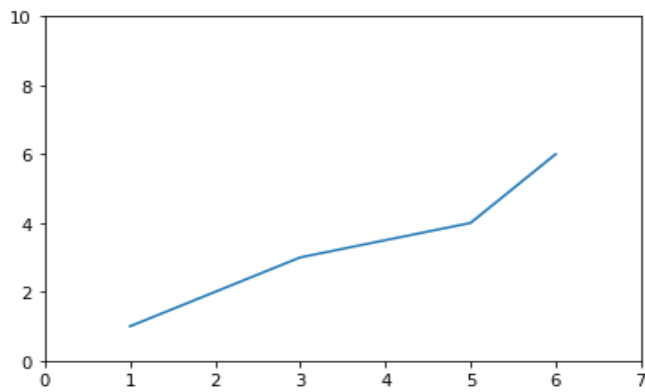
#### **(i) Creating of line plot**

##### **Program**

```
import matplotlib.pyplot as plt
plt.plot([1,2,3,5,6], [1, 2, 3, 4, 6])
plt.axis([0, 7, 0, 10])
```

```
plt.show()
```

### Output

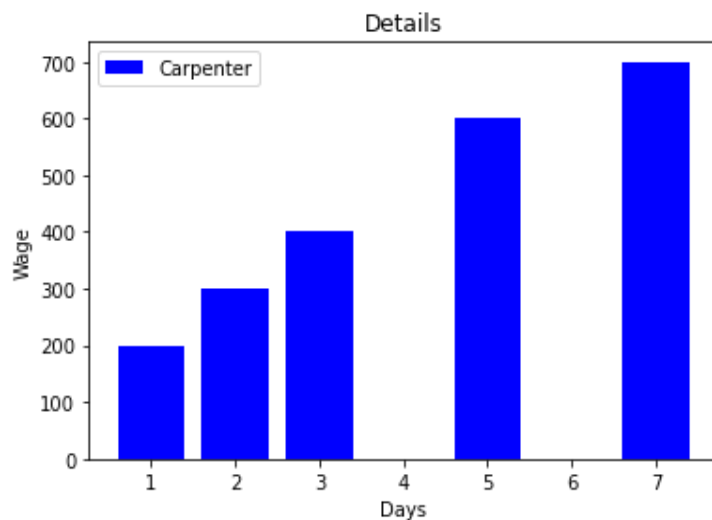


### (ii) Creating bar plot

#### Program

```
plt.bar([2,3,1,5,7],[300,400,200,600,700],  
label="Carpenter",color='b')  
plt.legend()  
plt.xlabel('Days')  
plt.ylabel('Wage')  
plt.title('Details')  
plt.show()
```

### Output



### (iii) Creating Scatter plot

#### Program

```
x1 = [1, 2.5,3,4.5,5,6.5,7]
y1 = [1,2, 3, 2, 1, 3, 4]
x2=[8, 8.5, 9, 9.5, 10, 10.5, 11]
y2=[3,3.5, 3.7, 4,4.5, 5, 5.2]

plt.scatter(x1, y1, label = 'high bp low heartrate', color='c')
plt.scatter(x2,y2,label='low bp high heartrate',color='g')

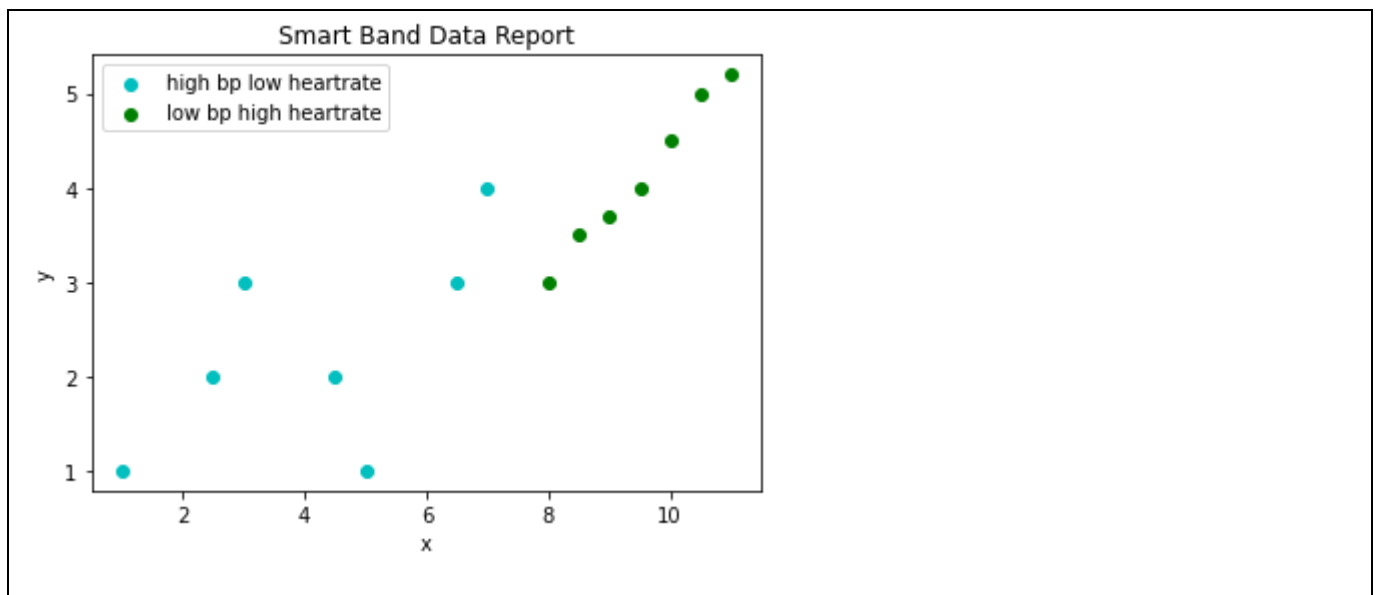
plt.title('Smart Band Data Report')

plt.xlabel('x')
plt.ylabel('y')

plt.legend()

plt.show()
```

#### Output



#### (iv) Creating Pie chart

##### Program

```
slice = [12, 25, 50, 36, 19]
```

```
activities = ['NLP', 'Neural Network', 'Data analytics', 'Quantum Computing', 'Machine Learning']
```

```
cols = ['r', 'b', 'c', 'g', 'orange']
```

```
plt.pie(slice,
```

```
labels = activities,
```

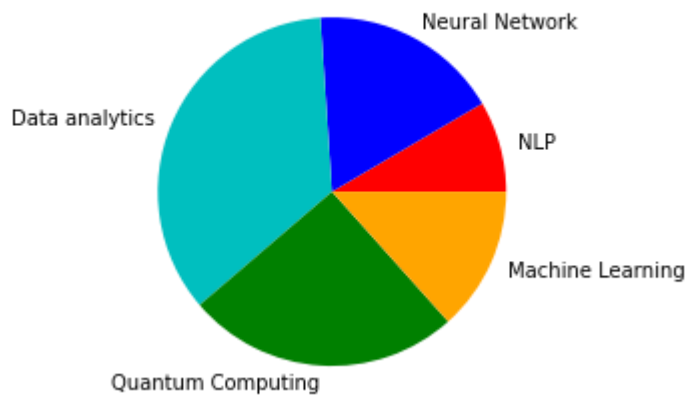
```
colors = cols,)
```

```
plt.title('Training Subjects')
```

```
plt.show()
```

##### Output

Training Subjects

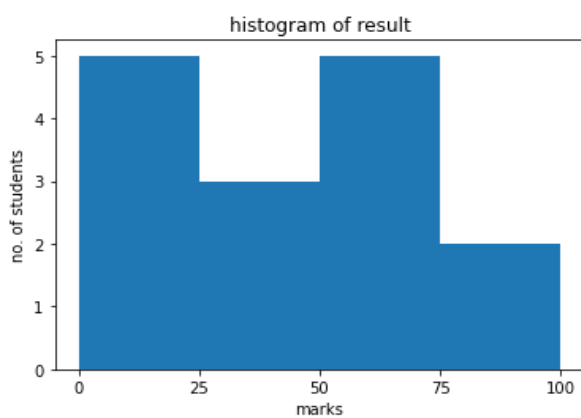


### (v) Creating Histogram plot

#### Program

```
fig,ax = plt.subplots(1,1)
a_new = np.array([22,87,5,43,56,73,55,54,11,20,51,5,79,31,27])
ax.hist(a_new, bins = [0,25,50,75,100])
ax.set_title("histogram of result")
ax.set_xticks([0,25,50,75,100])
ax.set_xlabel('marks')
ax.set_ylabel('no. of students')
plt.show()
```

#### Output



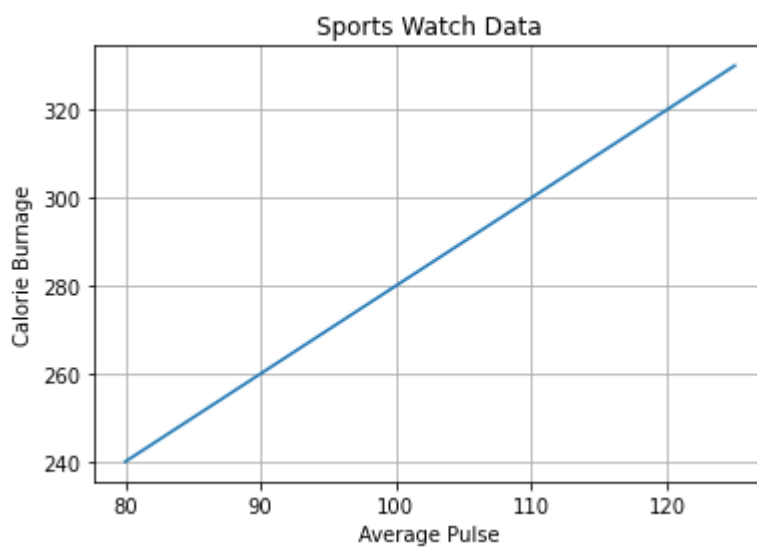
### (vi) Adding Gridlines to plot

#### Program

```
x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])

plt.title("Sports Watch Data")
plt.xlabel("Average Pulse")
plt.ylabel("Calorie Burnage")
plt.plot(x, y)
plt.grid()
plt.show()
```

#### Output



#### Result

Thus working with arrays in the numpy module, dataframe with pandas module and basic plots with matplotlib module in python programming has been explored successfully.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ex No : 4</b><br><br><b>DATE :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Explore various variable and row filters in R for cleaning data. Apply various plot features in R on sample data sets and visualize.</b> |
| <b>Aim:</b><br><br>The aim of this project is to explore data cleaning techniques in R, including variable and row filters, and to apply various plot features to sample datasets to visualize data effectively.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                             |
| <b>Algorithm:</b><br><br>Step 1: Import the required libraries for data manipulation, such as dplyr for filtering and cleaning, and ggplot2 for data visualization.<br><br>Step 2: Load one or more sample datasets for analysis. These datasets should have some missing values and outliers for data cleaning demonstrations.<br><br>Step 3: Apply variable filters (e.g., removing unnecessary columns) to clean the dataset.<br><br>Step 4: Apply row filters (e.g., removing rows with missing values or outliers) to clean the dataset further.<br><br>Step 5: Perform EDA to gain insights into the data.<br><br>Step 6: Create summary statistics, histograms, and box plots to understand the distribution of variables.<br><br>Step 7: Use ggplot2 to create various types of plots, such as bar charts, scatter plots, and line plots, to visualize the data.<br><br>Step 8: Display the created plots using R's built-in functionality.<br><br>Step 9: Optionally, save the plots as image files for further use or reporting. |                                                                                                                                             |
| <b>Program:</b><br><br># Load necessary libraries<br><br>library(dplyr)<br><br>library(ggplot2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                             |
| # Load sample dataset (e.g., 'mtcars')<br><br>data <- mtcars                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                             |
| # Data cleaning: Variable and row filters<br><br>cat("Original Data:\n")<br><br>print(data)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                             |

```
data_cleaned <- data %>%
```

```
  select(-c(vs, am)) %>%
```

```
  filter(!is.na(hp), hp >= 50, hp <= 300)
```

```
cat("\nData After Cleaning:\n")
```

```
print(data_cleaned)
```

```
# Exploratory Data Analysis (EDA)
```

```
cat("\nSummary of Cleaned Data:\n")
```

```
print(summary(data_cleaned))
```

```
hist(data_cleaned$mpg, main="Histogram of MPG", xlab="MPG")
```

```
boxplot(data_cleaned$hp, main="Boxplot of HP")
```

```
# Data visualization with ggplot2
```

```
cat("\nData Visualization:\n")
```

```
gg <- ggplot(data_cleaned, aes(x = wt, y = mpg)) +
```

```
  geom_point(aes(color = qsec)) +
```

```
  labs(title = "Scatter Plot of MPG vs. Weight",
```

```
        x = "Weight",
```

```
        y = "MPG")
```

```
print(g)
```

```
# Save the plot as an image file (optional)
```

```
ggsave("scatter_plot.png")
```

```
# Display the plot
```

```
print(g)
```

## Output:

Original Data:

|               | mpg  | cyl | disp  | hp  | drat | wt    | qsec  | vs | am | gear | carb |
|---------------|------|-----|-------|-----|------|-------|-------|----|----|------|------|
| Mazda RX4     | 21.0 | 6   | 160.0 | 110 | 3.90 | 2.620 | 16.46 | 0  | 1  | 4    | 4    |
| Mazda RX4 Wag | 21.0 | 6   | 160.0 | 110 | 3.90 | 2.875 | 17.02 | 0  | 1  | 4    | 4    |
| Datsun 710    | 22.8 | 4   | 108.0 | 93  | 3.85 | 2.320 | 18.61 | 1  | 1  | 4    | 1    |

|                      |      |     |       |     |      |       |       |      |      |   |   |
|----------------------|------|-----|-------|-----|------|-------|-------|------|------|---|---|
| Hornet 4 Drive       | 21.4 | 6   | 258.0 | 110 | 3.08 | 3.215 | 19.44 | 1    | 0    | 3 | 1 |
| Hornet Sportabout    | 18.7 | 8   | 360.0 | 175 | 3.15 | 3.440 | 17.02 | 0    | 0    | 3 | 2 |
| Valiant              | 18.1 | 6   | 225.0 | 105 | 2.76 | 3.460 | 20.22 | 1    | 0    | 3 | 1 |
| Duster 360           | 14.3 | 8   | 360.0 | 245 | 3.21 | 3.570 | 15.84 | 0    | 0    | 3 | 4 |
| Merc 240D            | 24.4 | 4   | 146.7 | 62  | 3.69 | 3.190 | 20.00 | 1    | 0    | 4 | 2 |
| Merc 230             | 22.8 | 4   | 140.8 | 95  | 3.92 | 3.150 | 22.90 | 1    | 0    | 4 | 2 |
| Merc 280             | 19.2 | 6   | 167.6 | 123 | 3.92 | 3.440 | 18.30 | 1    | 0    | 4 | 4 |
| Merc 280C            | 17.8 | 6   | 167.6 | 123 | 3.92 | 3.440 | 18.90 | 1    | 0    | 4 | 4 |
| Merc 450SE           | 16.4 | 8   | 275.8 | 180 | 3.07 | 4.070 | 17.40 | 0    | 0    | 3 | 3 |
| Merc 450SL           | 17.3 | 8   | 275.8 | 180 | 3.07 | 3.730 | 17.60 | 0    | 0    | 3 | 3 |
| Merc 450SLC          | 15.2 | 8   | 275.8 | 180 | 3.07 | 3.780 | 18.00 | 0    | 0    | 3 | 3 |
| Cadillac Fleetwood   | 10.4 | 8   | 472.0 | 205 | 2.93 | 5.250 | 17.98 | 0    | 0    | 3 | 4 |
| Lincoln Continental  | 10.4 | 8   | 460.0 | 215 | 3.00 | 5.424 | 17.82 | 0    | 0    | 3 | 4 |
| Chrysler Imperial    | 14.7 | 8   | 440.0 | 230 | 3.23 | 5.345 | 17.42 | 0    | 0    | 3 | 4 |
| Fiat 128             | 32.4 | 4   | 78.7  | 66  | 4.08 | 2.200 | 19.47 | 1    | 1    | 4 | 1 |
| Honda Civic          | 30.4 | 4   | 75.7  | 52  | 4.93 | 1.615 | 18.52 | 1    | 1    | 4 | 2 |
| Toyota Corolla       | 33.9 | 4   | 71.1  | 65  | 4.22 | 1.835 | 19.90 | 1    | 1    | 4 | 1 |
| Toyota Corona        | 21.5 | 4   | 120.1 | 97  | 3.70 | 2.465 | 20.01 | 1    | 0    | 3 | 1 |
| Dodge Challenger     | 15.5 | 8   | 318.0 | 150 | 2.76 | 3.520 | 16.87 | 0    | 0    | 3 | 2 |
| AMC Javelin          | 15.2 | 8   | 304.0 | 150 | 3.15 | 3.435 | 17.30 | 0    | 0    | 3 | 2 |
| Camaro Z28           | 13.3 | 8   | 350.0 | 245 | 3.73 | 3.840 | 15.41 | 0    | 0    | 3 | 4 |
| Pontiac Firebird     | 19.2 | 8   | 400.0 | 175 | 3.08 | 3.845 | 17.05 | 0    | 0    | 3 | 2 |
| Fiat X1-9            | 27.3 | 4   | 79.0  | 66  | 4.08 | 1.935 | 18.90 | 1    | 1    | 4 | 1 |
| Porsche 914-2        | 26.0 | 4   | 120.3 | 91  | 4.43 | 2.140 | 16.70 | 0    | 1    | 5 | 2 |
| Lotus Europa         | 30.4 | 4   | 95.1  | 113 | 3.77 | 1.513 | 16.90 | 1    | 1    | 5 | 2 |
| Ford Pantera L       | 15.8 | 8   | 351.0 | 264 | 4.22 | 3.170 | 14.50 | 0    | 1    | 5 | 4 |
| Ferrari Dino         | 19.7 | 6   | 145.0 | 175 | 3.62 | 2.770 | 15.50 | 0    | 1    | 5 | 6 |
| Maserati Bora        | 15.0 | 8   | 301.0 | 335 | 3.54 | 3.570 | 14.60 | 0    | 1    | 5 | 8 |
| Volvo 142E           | 21.4 | 4   | 121.0 | 109 | 4.11 | 2.780 | 18.60 | 1    | 1    | 4 | 2 |
| Data After Cleaning: |      |     |       |     |      |       |       |      |      |   |   |
|                      | mpg  | cyl | disp  | hp  | drat | wt    | qsec  | gear | carb |   |   |
| Mazda RX4            | 21.0 | 6   | 160.0 | 110 | 3.90 | 2.620 | 16.46 | 4    | 4    |   |   |
| Mazda RX4 Wag        | 21.0 | 6   | 160.0 | 110 | 3.90 | 2.875 | 17.02 | 4    | 4    |   |   |
| Datsun 710           | 22.8 | 4   | 108.0 | 93  | 3.85 | 2.320 | 18.61 | 4    | 1    |   |   |
| Hornet 4 Drive       | 21.4 | 6   | 258.0 | 110 | 3.08 | 3.215 | 19.44 | 3    | 1    |   |   |
| Hornet Sportabout    | 18.7 | 8   | 360.0 | 175 | 3.15 | 3.440 | 17.02 | 3    | 2    |   |   |
| Valiant              | 18.1 | 6   | 225.0 | 105 | 2.76 | 3.460 | 20.22 | 3    | 1    |   |   |
| Duster 360           | 14.3 | 8   | 360.0 | 245 | 3.21 | 3.570 | 15.84 | 3    | 4    |   |   |
| Merc 240D            | 24.4 | 4   | 146.7 | 62  | 3.69 | 3.190 | 20.00 | 4    | 2    |   |   |
| Merc 230             | 22.8 | 4   | 140.8 | 95  | 3.92 | 3.150 | 22.90 | 4    | 2    |   |   |
| Merc 280             | 19.2 | 6   | 167.6 | 123 | 3.92 | 3.440 | 18.30 | 4    | 4    |   |   |
| Merc 280C            | 17.8 | 6   | 167.6 | 123 | 3.92 | 3.440 | 18.90 | 4    | 4    |   |   |
| Merc 450SE           | 16.4 | 8   | 275.8 | 180 | 3.07 | 4.070 | 17.40 | 3    | 3    |   |   |
| Merc 450SL           | 17.3 | 8   | 275.8 | 180 | 3.07 | 3.730 | 17.60 | 3    | 3    |   |   |
| Merc 450SLC          | 15.2 | 8   | 275.8 | 180 | 3.07 | 3.780 | 18.00 | 3    | 3    |   |   |
| Cadillac Fleetwood   | 10.4 | 8   | 472.0 | 205 | 2.93 | 5.250 | 17.98 | 3    | 4    |   |   |

|                     |      |   |       |     |      |       |       |   |   |
|---------------------|------|---|-------|-----|------|-------|-------|---|---|
| Lincoln Continental | 10.4 | 8 | 460.0 | 215 | 3.00 | 5.424 | 17.82 | 3 | 4 |
| Chrysler Imperial   | 14.7 | 8 | 440.0 | 230 | 3.23 | 5.345 | 17.42 | 3 | 4 |
| Fiat 128            | 32.4 | 4 | 78.7  | 66  | 4.08 | 2.200 | 19.47 | 4 | 1 |
| Honda Civic         | 30.4 | 4 | 75.7  | 52  | 4.93 | 1.615 | 18.52 | 4 | 2 |
| Toyota Corolla      | 33.9 | 4 | 71.1  | 65  | 4.22 | 1.835 | 19.90 | 4 | 1 |
| Toyota Corona       | 21.5 | 4 | 120.1 | 97  | 3.70 | 2.465 | 20.01 | 3 | 1 |
| Dodge Challenger    | 15.5 | 8 | 318.0 | 150 | 2.76 | 3.520 | 16.87 | 3 | 2 |
| AMC Javelin         | 15.2 | 8 | 304.0 | 150 | 3.15 | 3.435 | 17.30 | 3 | 2 |
| Camaro Z28          | 13.3 | 8 | 350.0 | 245 | 3.73 | 3.840 | 15.41 | 3 | 4 |
| Pontiac Firebird    | 19.2 | 8 | 400.0 | 175 | 3.08 | 3.845 | 17.05 | 3 | 2 |
| Fiat X1-9           | 27.3 | 4 | 79.0  | 66  | 4.08 | 1.935 | 18.90 | 4 | 1 |
| Porsche 914-2       | 26.0 | 4 | 120.3 | 91  | 4.43 | 2.140 | 16.70 | 5 | 2 |
| Lotus Europa        | 30.4 | 4 | 95.1  | 113 | 3.77 | 1.513 | 16.90 | 5 | 2 |
| Ford Pantera L      | 15.8 | 8 | 351.0 | 264 | 4.22 | 3.170 | 14.50 | 5 | 4 |
| Ferrari Dino        | 19.7 | 6 | 145.0 | 175 | 3.62 | 2.770 | 15.50 | 5 | 6 |
| Volvo 142E          | 21.4 | 4 | 121.0 | 109 | 4.11 | 2.780 | 18.60 | 4 | 2 |

Summary of Cleaned Data:

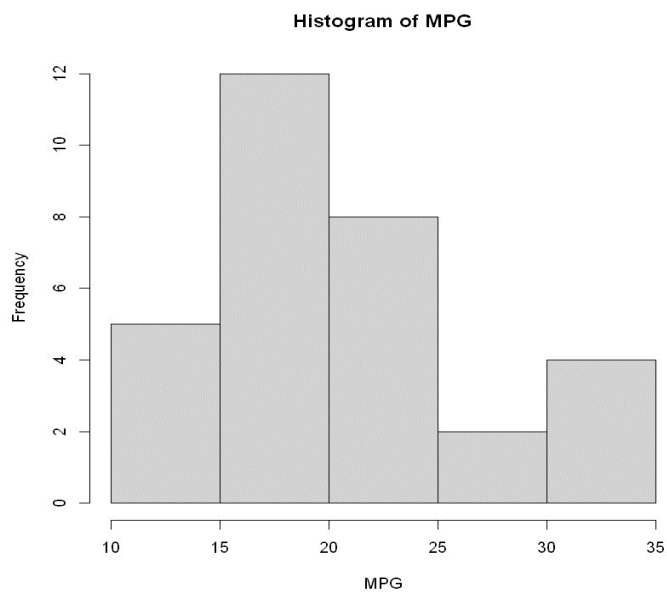
| mpg           | cyl           | disp          | hp            |
|---------------|---------------|---------------|---------------|
| Min. :10.40   | Min. :4.000   | Min. : 71.1   | Min. : 52.0   |
| 1st Qu.:15.65 | 1st Qu.:4.000 | 1st Qu.:120.7 | 1st Qu.: 96.0 |
| Median :19.20 | Median :6.000 | Median :167.6 | Median :123.0 |
| Mean :20.25   | Mean :6.129   | Mean :228.5   | Mean :140.6   |
| 3rd Qu.:22.80 | 3rd Qu.:8.000 | 3rd Qu.:334.0 | 3rd Qu.:180.0 |
| Max. :33.90   | Max. :8.000   | Max. :472.0   | Max. :264.0   |

| drat          | wt            | qsec          | gear          |
|---------------|---------------|---------------|---------------|
| Min. :2.760   | Min. :1.513   | Min. :14.50   | Min. :3.000   |
| 1st Qu.:3.080 | 1st Qu.:2.542 | 1st Qu.:16.96 | 1st Qu.:3.000 |
| Median :3.700 | Median :3.215 | Median :17.82 | Median :4.000 |
| Mean :3.598   | Mean :3.206   | Mean :17.95   | Mean :3.645   |
| 3rd Qu.:3.920 | 3rd Qu.:3.650 | 3rd Qu.:18.90 | 3rd Qu.:4.000 |
| Max. :4.930   | Max. :5.424   | Max. :22.90   | Max. :5.000   |

| carb          |
|---------------|
| Min. :1.000   |
| 1st Qu.:2.000 |
| Median :2.000 |
| Mean :2.645   |
| 3rd Qu.:4.000 |
| Max. :6.000   |



**Result:**

Thus the process of data manipulation and visualization using R was executed successfully.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Ex No : 5</b><br><br><b>DATE :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Perform Time Series Analysis and apply the various visualization techniques</b> |
| <b>Aim :</b><br><br>To Write a code to implement the Time Series Analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                    |
| <b>Algorithm :</b><br><br>Step 1: Import the required libraries: <ol style="list-style-type: none"> <li>1. Pandas for data handling.</li> <li>2. Numpy for mathematical processing.</li> <li>3. Matplotlib for data visualization.</li> <li>4. Statsmodels for the time series</li> </ol> Step 2: Set the random seed for reproducibility using <code>np.random.seed ( )</code> .<br>Step 3: Set the date range using <code>pd.date_range( )</code> .<br>Step 4: Generate the random values for the date from 1 to 100 using <code>np.random.randint ( )</code> and convert it into a Data Frame.<br>Step 5: Set the Date as index to the DataFrame using <code>df.set_index ( 'date' , inplace=True)</code><br>Step 6: From the statsmodels import <code>seasonal_decompose</code> to show different time series visuals like trend , Seasonal , resid etc ...<br>Step 7: Visualize the DataFrame using Matplot library as <code>df.plot( )</code> .<br>Step 8: Create a variable decomposition and pass the model as <code>seasonal_decompose( df['value'],model='Additive')</code> and use the variable by accesing all the visuals ( <code>decomposition.trend,</code><br><code>decomposition.Seasonal,decomposition.reside)</code><br>Step 9: Finally, use <code>plt.show( )</code> to show the Visualization. |                                                                                    |
| <b>Program :</b><br><br><pre> import pandas as pd import numpy as np import matplotlib.pyplot as plt </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                    |

```

from statsmodels.tsa.seasonal import seasonal_decompose
np.random.seed(0)
date_range = pd.date_range('2023-01-01','2023-12-31',freq='D')
data = np.random.randint(low=1,high=100,size=len(date_range))
df = pd.DataFrame({'Date':date_range,'Value':data})
df.set_index('Date',inplace=True)
df.head()

```

**Output:**

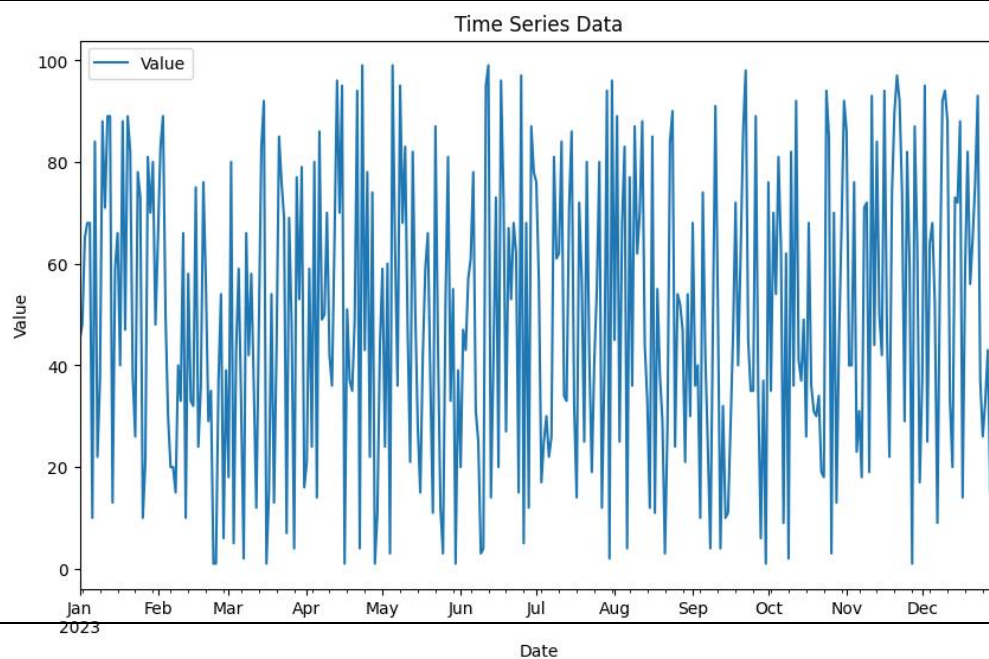
| Date       | Value |
|------------|-------|
| 2023-01-01 | 45    |
| 2023-01-02 | 48    |
| 2023-01-03 | 65    |
| 2023-01-04 | 68    |
| 2023-01-05 | 68    |

```

df.plot(figsize=(10, 6))
plt.xlabel('Date')
plt.ylabel('Value')
plt.title('Time Series Data')
plt.show()

```

**Output:**



```
decomposition = seasonal_decompose(df['Value'], model='additive')

trend = decomposition.trend

seasonality = decomposition.seasonal

residuals = decomposition.resid

plt.figure(figsize=(12, 8))

plt.subplot(411)

plt.plot(df['Value'], label='Original')

plt.legend(loc='best')

plt.subplot(412)

plt.plot(trend, label='Trend')

plt.legend(loc='best')

plt.subplot(413)

plt.plot(seasonality, label='Seasonality')

plt.legend(loc='best')

plt.subplot(414)

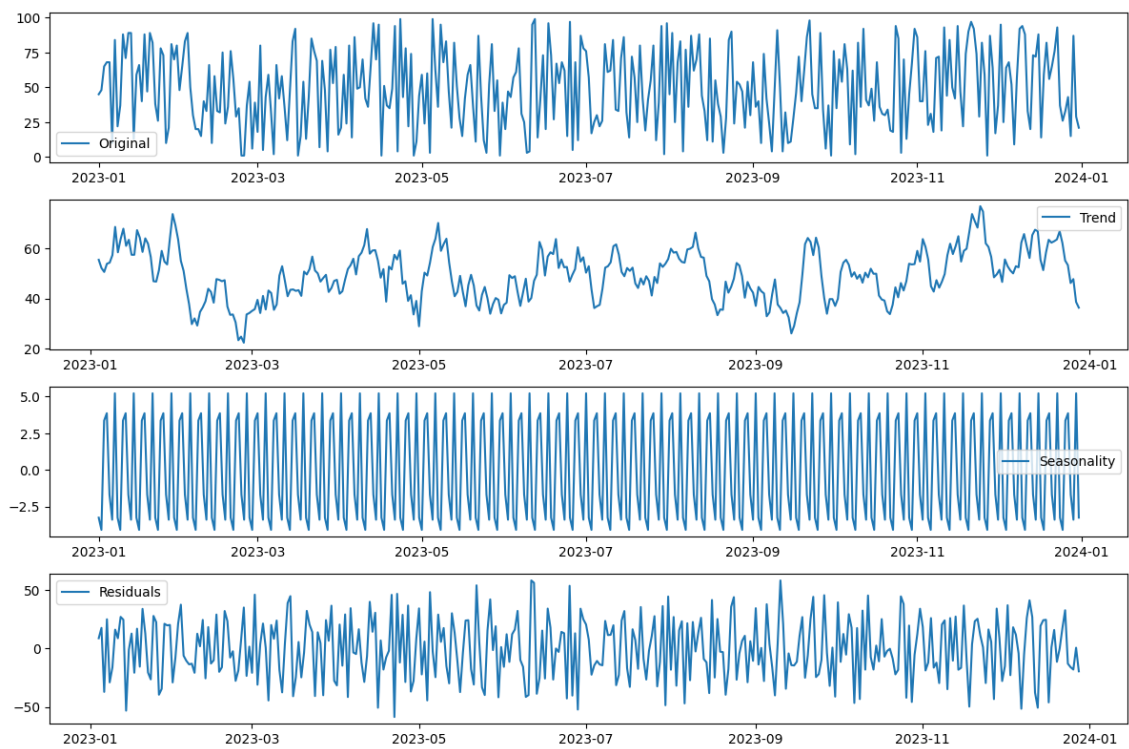
plt.plot(residuals, label='Residuals')

plt.legend(loc='best')

plt.tight_layout()

plt.show()
```

**Output:**



**RESULT :**

Thus, the above Times Series program was written and executed successfully.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ex No : 6</b><br><br><b>Date :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Perform data analysis and representation on a map using various map datasets with mouse rollover effect, user interaction, etc.,</b> |
| <b>Aim:</b><br><br>To Perform data analysis and representation on a map using various map datasets with mouse rollover effect, user interaction, etc,.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                         |
| <b>Algorithm:</b><br><br><p><b>Step 1:</b> Collect and prepare various map datasets, including geographical information (latitude, longitude), and relevant data attributes.</p> <p><b>Step 2:</b> Choose a suitable map library or framework (e.g., Leaflet, Google Maps API) for displaying the maps.</p> <p><b>Step 3:</b> Create a web-based application that renders the map and overlays it with markers, polygons, or other visual elements based on the dataset.</p> <p><b>Step 4:</b> Implement a mouse rollover effect, where users can hover over map elements (markers, regions) to view additional information related to the data point.</p> <p><b>Step 5:</b> Enable user interaction by allowing users to interact with the map, such as zooming in/out, panning, and selecting specific data points for more details.</p> <p><b>Step 6:</b> Perform data analysis on the selected dataset, which may include generating statistics, clustering, or creating heatmaps based on geographical attributes.</p> <p><b>Step 7:</b> Implement filters and visualization options, such as color-coding or varying marker sizes to represent different data attributes.</p> <p><b>Step 8:</b> Design a user-friendly interface with clear controls, legends, and tooltips to help users understand the data and its representation on the map.</p> |                                                                                                                                         |
| <b>Program:</b><br><br><pre> # Import necessary libraries or frameworks  import map_library  import data_processing  # Load map data  map_data = data_processing.load_map_data()  # Initialize the map </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                         |

```
map = map_library.initialize_map()

# Add data markers and polygons to the map

for data_point in map_data:

    map.add_marker(data_point)

# Implement mouse rollover effect

map.enable_rollover_effect()

# Implement user interaction (e.g., zoom, pan)

map.enable_user_interaction()

# Perform data analysis (e.g., calculate statistics, clusters)

data_analysis = data_processing.analyze_data(map_data)

# Visualize data on the map (e.g., color-coded markers)

map.visualize_data(data_analysis)

# Create a user-friendly interface with controls and legends

map.create_user_interface()

# Display the map and interface

map.show_map()
```

### Output:



**Result:**

The web-based interactive map application that allows users to explore geographical data using mouse rollover effects, interactive features has been created.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ex No : 7</b><br><br><b>Date :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Build cartographic visualisation of multiple datasets involving various countries of the world; states, and districts in India</b> |
| <b>Aim:</b><br><br>To Build cartographic visualisation of multiple datasets involving various countries of the world; states, and districts in India etc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                       |
| <b>Algorithm:</b><br><br><b>Step 1:</b> Gather the multiple datasets related to countries, states, and districts, and ensure that each dataset contains geographical information, such as latitude and longitude, to accurately plot the data on maps.<br><br><b>Step 2:</b> Choose a mapping library or tool suitable for the project, such as Leaflet, Mapbox, or Google Maps.<br><br><b>Step 3:</b> Develop a web-based application or interactive dashboard that renders the world map and India's map. Overlay the map with relevant boundaries (country borders, state borders, district boundaries).<br><br><b>Step 4:</b> Merge the datasets with the map layers, linking data points to their geographical locations (countries, states, districts).<br><br><b>Step 5:</b> Implement data visualization techniques to represent the datasets visually on the maps. This may include choropleth maps, bubble maps, or heatmaps, depending on the nature of the data.<br><br><b>Step 6:</b> Apply color-coding to the map elements to represent different data attributes, making it easy to distinguish and analyze the information.<br><br><b>Step 7:</b> Enable user interaction by allowing users to zoom in/out, pan, and click on map elements to access detailed information about the regions or data points.<br><br><b>Step 8:</b> Include legends, labels, and tooltips to help users interpret the visualizations and understand the meaning of different colors and symbols. |                                                                                                                                       |

**Program:**

```
# Import necessary libraries or mapping tools

import mapping_library

import data_processing

# Load geographical data (countries, states, districts)

world_map_data = data_processing.load_world_map_data()

india_map_data = data_processing.load_india_map_data()

# Initialize the maps

world_map = mapping_library.initialize_map()

india_map = mapping_library.initialize_map()

# Add map layers (country, state, district boundaries)

world_map.add_layers(world_map_data)

india_map.add_layers(india_map_data)

# Merge datasets with map layers

world_data = data_processing.merge_data_with_map(world_map, world_map_data)

india_data = data_processing.merge_data_with_map(india_map, india_map_data)

# Visualize data on maps (e.g., choropleth, bubble maps)

world_map.visualize_data(world_data)

india_map.visualize_data(india_data)

# Apply color-coding to represent data attributes

world_map.apply_color_coding()

india_map.apply_color_coding()

# Implement user interaction (zoom, pan, click)

world_map.enable_user_interaction()

india_map.enable_user_interaction()
```

```
# Create legends and tooltips for data interpretation
```

```
world_map.create_legends_and_labels()
```

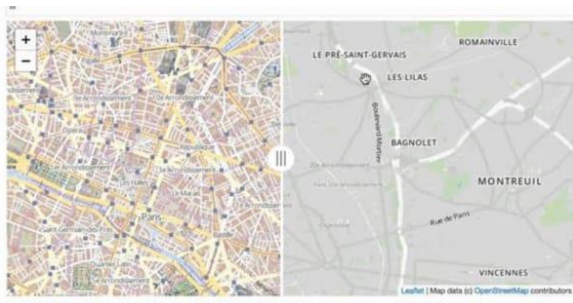
```
india_map.create_legends_and_labels()
```

```
# Display the maps and interface
```

```
world_map.show_map()
```

```
india_map.show_map()
```

### Output:



### Result:

The result is an interactive web-based platform or application that provides cartographic visualizations for multiple datasets related to countries worldwide, as well as states and districts in India.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EX NO: 8                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Perform EDA on Wine Quality Data Set |
| DATE:                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                      |
| <b>Aim</b><br><br>To perform EDA on Wine quality dataset.                                                                                                                                                                                                                                                                                                                                                                                                             |                                      |
| <b>Algorithm</b><br><br><b>Step 1:</b> Import necessary libraries such as pandas, numpy, matplotlib and seaborn.<br><br><b>Step 2:</b> Read a CSV file ('winequality-red.csv') into a Pandas DataFrame and store it in the variable<br><br><b>Step 3:</b> Take the first column of the DataFrame and split it into multiple columns based on the delimiter ';'<br><br><b>Step 4:</b> Change the data types of the columns in the DataFrame to their appropriate types |                                      |

**Step 5:** Check the datatypes, missing values, and summary of the data.

**Step 6:** Create a histogram of the 'quality' column using Seaborn with labels and a title, and display the plot.

**Step 7:** Calculate the correlation matrix for the dataset and create a heatmap using Seaborn with annotations and a title, and display the plot.

**Step 8:** Create a scatter plot of 'alcohol' vs. 'quality' using Seaborn with labels and a title, and display the plot.

**Program :**

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

data = pd.read_csv('/content/sample_data/winequality-red.csv')

print(data.columns)
```

**Output:**

```
Index(['fixed acidity', 'volatile acidity', 'citric acid', 'residual sugar',
      'chlorides', 'free sulfur dioxide', 'total sulfur dioxide', 'density',
      'pH', 'sulphates', 'alcohol', 'quality'],
      dtype='object')
```

**# Split the single column into multiple columns based on the delimiter (;)**

```
data = data[data.columns[0]].str.split(';', expand=True)
```

**# Rename the columns based on the original column names**

```
column_names = ['fixed acidity', 'volatile acidity', 'citric acid', 'residual sugar', 'chlorides',
                'free sulfur dioxide', 'total sulfur dioxide', 'density', 'pH', 'sulphates',
                'alcohol', 'quality']

data.columns = column_names
```

**# Convert the columns to appropriate data types**

```
data = data.astype({'fixed acidity': float, 'volatile acidity': float, 'citric acid': float,
                   'residual sugar': float, 'chlorides': float, 'free sulfur dioxide': float,
```

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| 'total sulfur dioxide': float, 'density': float, 'pH': float,<br>'sulphates': float, 'alcohol': float, 'quality': int})                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |  |  |  |
| <b># View the first few rows of the dataset</b><br><br>print(data.head())                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |  |  |  |
| <b>Output</b><br><br>fixed acidity volatile acidity citric acid residual sugar chlorides \<br>0 7.4 0.70 0.00 1.9 0.076<br>1 7.8 0.88 0.00 2.6 0.098<br>2 7.8 0.76 0.04 2.3 0.092<br>3 11.2 0.28 0.56 1.9 0.075<br>4 7.4 0.70 0.00 1.9 0.076<br><br>free sulfur dioxide total sulfur dioxide density pH sulphates \<br>0 11.0 34.0 0.9978 3.51 0.56<br>1 25.0 67.0 0.9968 3.20 0.68<br>2 15.0 54.0 0.9970 3.26 0.65<br>3 17.0 60.0 0.9980 3.16 0.58<br>4 11.0 34.0 0.9978 3.51 0.56<br><br>alcohol quality<br>0 9.4 5<br>1 9.8 5<br>2 9.8 5<br>3 9.8 6<br>4 9.4 5 |  |  |  |  |  |
| <b># Check the dimensions of the dataset</b><br><br>print(data.shape)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |  |  |  |

**Output**

(1599, 12)

**# Check the data types of each column**

```
print(data.dtypes)
```

**Output**

```
fixed acidity      float64
volatile acidity   float64
citric acid        float64
residual sugar     float64
chlorides          float64
free sulfur dioxide float64
total sulfur dioxide float64
density           float64
pH               float64
sulphates         float64
alcohol           float64
quality           int64
dtype: object
```

**# Check for missing values**

```
print(data.isnull().sum())
```

**Output**

```
fixed acidity      0
volatile acidity    0
citric acid        0
residual sugar     0
```

```

chlorides      0
free sulfur dioxide  0
total sulfur dioxide  0
density        0
pH             0
sulphates      0
alcohol        0
quality        0
dtype: int64

```

```
# Generate summary statistics
```

```
print(data.describe())
```

### Output

```

      fixed acidity  volatile acidity  citric acid  residual sugar \
count  1599.000000    1599.000000  1599.000000    1599.000000
mean     8.319637      0.527821    0.270976      2.538806
std     1.741096      0.179060    0.194801      1.409928
min      4.600000      0.120000    0.000000      0.900000
25%      7.100000      0.390000    0.090000      1.900000
50%      7.900000      0.520000    0.260000      2.200000
75%      9.200000      0.640000    0.420000      2.600000
max     15.900000      1.580000    1.000000     15.500000

      chlorides  free sulfur dioxide  total sulfur dioxide   density \
count  1599.000000    1599.000000    1599.000000  1599.000000
mean    0.087467     15.874922     46.467792   0.996747
std     0.047065     10.460157     32.895324   0.001887

```

|     |          |           |            |          |
|-----|----------|-----------|------------|----------|
| min | 0.012000 | 1.000000  | 6.000000   | 0.990070 |
| 25% | 0.070000 | 7.000000  | 22.000000  | 0.995600 |
| 50% | 0.079000 | 14.000000 | 38.000000  | 0.996750 |
| 75% | 0.090000 | 21.000000 | 62.000000  | 0.997835 |
| max | 0.611000 | 72.000000 | 289.000000 | 1.003690 |

|  |    |           |         |         |
|--|----|-----------|---------|---------|
|  | pH | sulphates | alcohol | quality |
|--|----|-----------|---------|---------|

|       |             |             |             |             |
|-------|-------------|-------------|-------------|-------------|
| count | 1599.000000 | 1599.000000 | 1599.000000 | 1599.000000 |
|-------|-------------|-------------|-------------|-------------|

|      |          |          |           |          |
|------|----------|----------|-----------|----------|
| mean | 3.311113 | 0.658149 | 10.422983 | 5.636023 |
|------|----------|----------|-----------|----------|

|     |          |          |          |          |
|-----|----------|----------|----------|----------|
| std | 0.154386 | 0.169507 | 1.065668 | 0.807569 |
|-----|----------|----------|----------|----------|

|     |          |          |          |          |
|-----|----------|----------|----------|----------|
| min | 2.740000 | 0.330000 | 8.400000 | 3.000000 |
|-----|----------|----------|----------|----------|

|     |          |          |          |          |
|-----|----------|----------|----------|----------|
| 25% | 3.210000 | 0.550000 | 9.500000 | 5.000000 |
|-----|----------|----------|----------|----------|

|     |          |          |           |          |
|-----|----------|----------|-----------|----------|
| 50% | 3.310000 | 0.620000 | 10.200000 | 6.000000 |
|-----|----------|----------|-----------|----------|

|     |          |          |           |          |
|-----|----------|----------|-----------|----------|
| 75% | 3.400000 | 0.730000 | 11.100000 | 6.000000 |
|-----|----------|----------|-----------|----------|

|     |          |          |           |          |
|-----|----------|----------|-----------|----------|
| max | 4.010000 | 2.000000 | 14.900000 | 8.000000 |
|-----|----------|----------|-----------|----------|

### # Histogram of wine quality

```
sns.histplot(data['quality'])
```

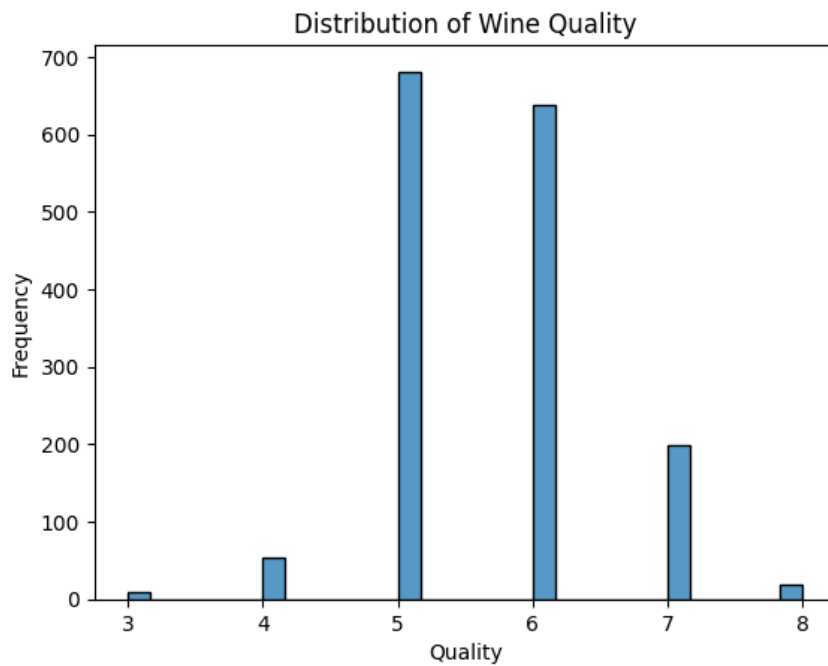
```
plt.xlabel('Quality')
```

```
plt.ylabel('Frequency')
```

```
plt.title('Distribution of Wine Quality')
```

```
plt.show()
```

### Output



### # Correlation matrix heatmap

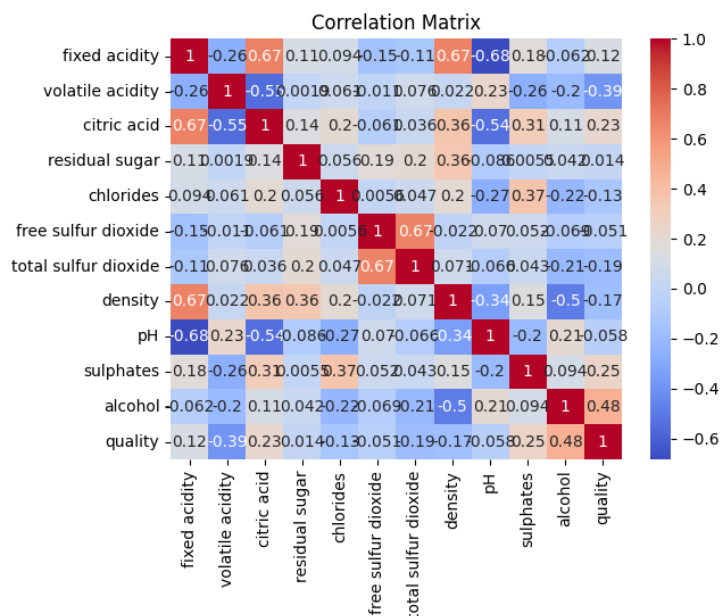
```
correlation_matrix = data.corr()

sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')

plt.title('Correlation Matrix')

plt.show()
```

### Output



### # Scatter plot of alcohol vs. quality

```
sns.scatterplot(x='alcohol', y='quality', data=data)

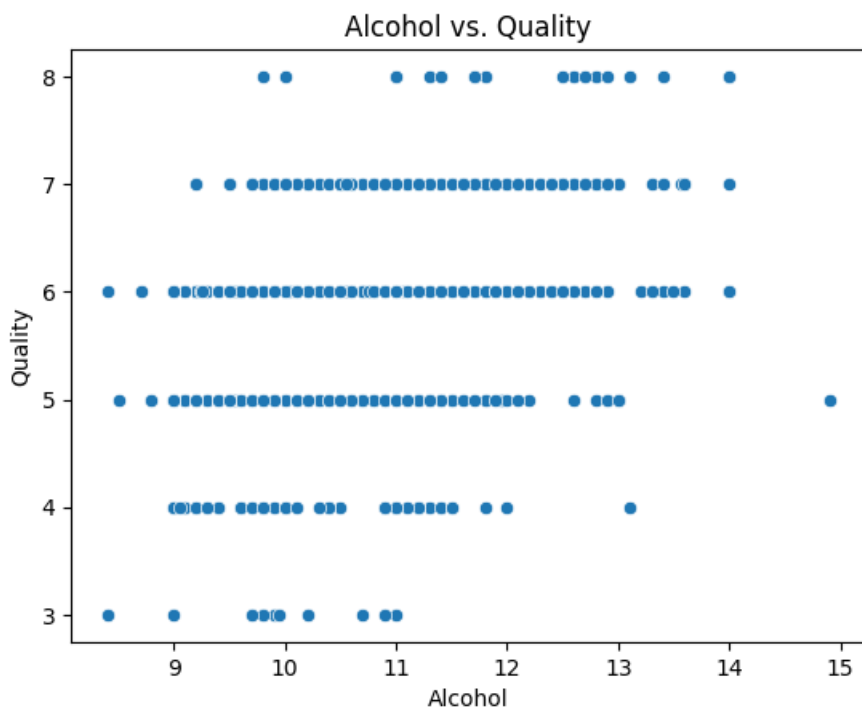
plt.xlabel('Alcohol')

plt.ylabel('Quality')

plt.title('Alcohol vs. Quality')

plt.show()
```

### Output



### Result :

Thus EDA on Wine quality dataset has been performed successfully.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Ex No : 9</b><br><br><b>Date :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Use a case study on a dataset and apply the various EDA and visualization techniques and present an analysis report</b> |
| <b>Aim:</b><br><br>To use a case study on a dataset and apply the various EDA and visualization techniques and present an analysis report                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                            |
| <b>Algorithm :</b><br><br><b>Step 1:</b> Import the required libraries.<br><br><b>Step 2:</b> Load the dataset.<br><br><b>Step 3:</b> Display basic information about the dataset.<br><br><b>Step 4:</b> Check for missing values.<br><br><b>Step 5:</b> Visualise the distribution of scores using various plots.<br><br><b>Step 6:</b> Check the correlation for variables.<br><br><b>Step 7:</b> Visualise the correlation using heatmap.<br><br><b>Step 8:</b> Give an overall analysis report. |                                                                                                                            |

**Program:**

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from scipy import stats

import statsmodels.api as sms

%matplotlib inline
```

```
df = pd.read_csv('StudentsPerformance.csv')
```

```
df
```

|     | gender | race/ethnicity | parental level of education | lunch        | test preparation course | math score | reading score | writing score |
|-----|--------|----------------|-----------------------------|--------------|-------------------------|------------|---------------|---------------|
| 0   | female | group B        | bachelor's degree           | standard     | none                    | 72         | 72            | 74            |
| 1   | female | group C        | some college                | standard     | completed               | 69         | 90            | 88            |
| 2   | female | group B        | master's degree             | standard     | none                    | 90         | 95            | 93            |
| 3   | male   | group A        | associate's degree          | free/reduced | none                    | 47         | 57            | 44            |
| 4   | male   | group C        | some college                | standard     | none                    | 76         | 78            | 75            |
| ... | ...    | ...            | ...                         | ...          | ...                     | ...        | ...           | ...           |
| 995 | female | group E        | master's degree             | standard     | completed               | 88         | 99            | 95            |
| 996 | male   | group C        | high school                 | free/reduced | none                    | 62         | 55            | 55            |
| 997 | female | group C        | high school                 | free/reduced | completed               | 59         | 71            | 65            |
| 998 | female | group D        | some college                | standard     | completed               | 68         | 78            | 77            |
| 999 | female | group D        | some college                | free/reduced | none                    | 77         | 86            | 86            |

1000 rows × 8 columns

```
df.info()
```

**Output :**

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1000 entries, 0 to 999
```

```
Data columns (total 8 columns):
```

| # | Column                      | Non-Null Count | Dtype  |
|---|-----------------------------|----------------|--------|
| 0 | gender                      | 1000 non-null  | object |
| 1 | race/ethnicity              | 1000 non-null  | object |
| 2 | parental level of education | 1000 non-null  | object |
| 3 | lunch                       | 1000 non-null  | object |
| 4 | test preparation course     | 1000 non-null  | object |
| 5 | math score                  | 1000 non-null  | int64  |
| 6 | reading score               | 1000 non-null  | int64  |
| 7 | writing score               | 1000 non-null  | int64  |

```
dtypes: int64(3), object(5)
```

```
memory usage: 62.6+ KB
```

```
df.isna().sum()
```

**Output :**

```
gender                0
race/ethnicity        0
parental level of education  0
lunch                0
test preparation course  0
math score            0
reading score         0
writing score         0
dtype: int64
```

```
df.describe()
```

**Output :**

|       | math score | reading score | writing score |
|-------|------------|---------------|---------------|
| count | 1000.00000 | 1000.000000   | 1000.000000   |
| mean  | 66.08900   | 69.169000     | 68.054000     |
| std   | 15.16308   | 14.600192     | 15.195657     |
| min   | 0.00000    | 17.000000     | 10.000000     |
| 25%   | 57.00000   | 59.000000     | 57.750000     |
| 50%   | 66.00000   | 70.000000     | 69.000000     |
| 75%   | 77.00000   | 79.000000     | 79.000000     |
| max   | 100.00000  | 100.000000    | 100.000000    |

```
sns.set_style('whitegrid')
```

```
plt.figure(figsize=(10, 6))
```

```
sns.boxplot(data=[df['math score'], df['reading score'], df['writing score']])
```

```
plt.xticks([0, 1, 2], ['math', 'reading', 'writing'])
```

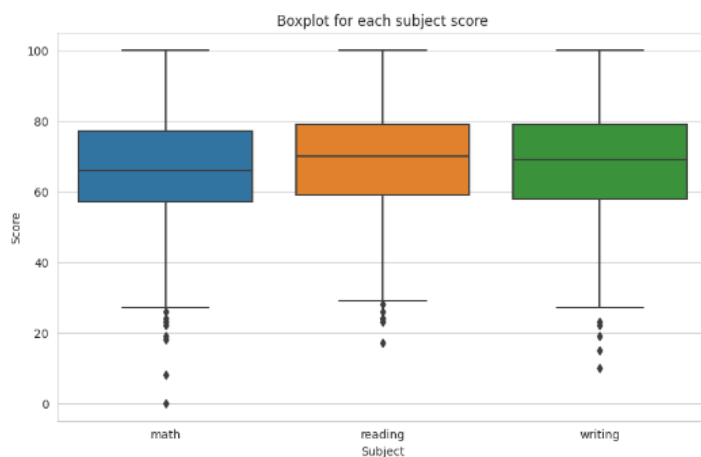
```
plt.xlabel('Subject')
```

```
plt.ylabel('Score')
```

```
plt.title('Boxplot for each subject score')
```

```
plt.show()
```

**Output :**



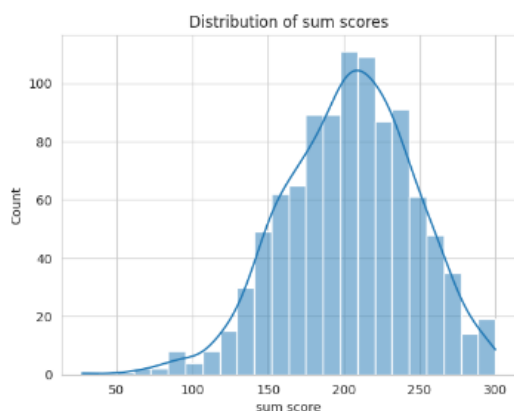
```
df['sum score'] = df[['math score', 'reading score', 'writing score']].sum(axis=1)

sns.histplot(df['sum score'], kde=True)

plt.title('Distribution of sum scores')

plt.show()
```

**Output :**



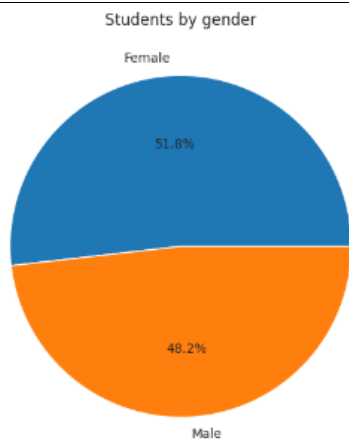
```
plt.figure(figsize=(10, 6))

plt.pie(df['gender'].value_counts(), autopct='%1.1f%%', labels=['Female', 'Male'])

plt.title('Students by gender')

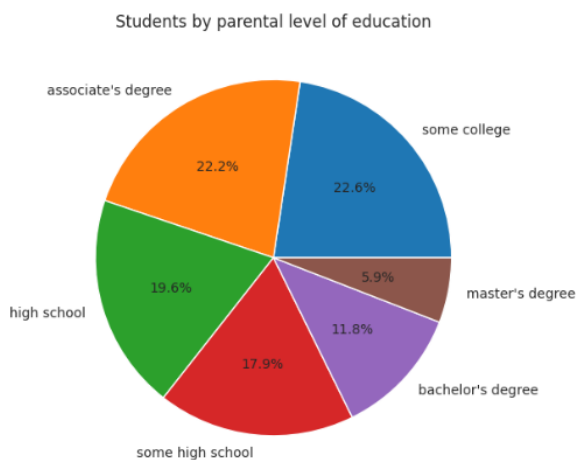
plt.show()
```

**Output :**



```
labels = df['parental level of education'].value_counts().index
plt.figure(figsize=(10, 6))
plt.pie(df['parental level of education'].value_counts(), autopct='%1.1f%%', labels=labels)
plt.title('Students by parental level of education')
plt.show()
```

**Output :**



```
plt.figure(figsize=(10, 6))
sns.heatmap(df_encoded.corr(), annot=True)
plt.show()
```

**Output :**

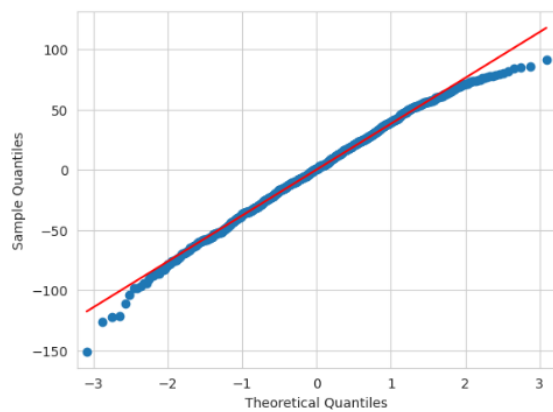


```
residuals = model.resid
```

```
sms.qqplot(residuals, line='s')
```

```
plt.show()
```

**Output :**



**Result :**

Thus by applying various EDA and visualization techniques we analysed the student score data and identified important factors affecting the total scores.

## EVALUATION SHEET

| <b>DEPARTMENT OF CSE</b> |                    |                   |                   |
|--------------------------|--------------------|-------------------|-------------------|
| <b>S.No.</b>             | <b>Description</b> | <b>Max. Marks</b> | <b>Min. Marks</b> |
| 1.                       | Aim/Procedure      | 30                |                   |
| 2.                       | Program            | 30                |                   |
| 3.                       | Output/Result      | 20                |                   |
| 4.                       | Viva Voce          | 10                |                   |
| 5.                       | Record             | 10                |                   |
| <b>Total:</b>            |                    | 100               |                   |